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Hidden alpha[☆]

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ABSTRACT

We provide novel evidence suggestive of insider trading through concealed relationships identified using information from over 100,000 Facebook profiles and their 35 million friends. Focusing on connections between fund managers and firm officers, we demonstrate that hidden ties are linked to substantial abnormal returns averaging 135 basis points per month (exceeding 16% alpha annually, t-stat = 3.54) across the universe of mutual funds and public firms. These hidden ties emerge as the most powerful predictor of future stock returns among documented network characteristics, with predictive power increasing over time through the present day. The premium associated with such connections arises not from endogenous selection or familiarity bias; instead, fund managers exhibit specific timing ability in deciding when to hold (or avoid) stocks of firm officers linked through hidden ties. The value of trading information rises with the degree of concealment and is concentrated around earnings and M&A events. The premium is absent in index funds, where strategic stock selection and timing are infeasible. Our findings on the value of hidden ties remain robust across industries, investment styles, time periods, and firm types.

Insider trading remains a central concern in securities markets and continues to attract substantial attention from regulators and academics alike. This interest is reflected in the U.S. Securities and Exchange Commission's (SEC's) ongoing expansion of both the scope of its oversight and the size of its enforcement workforce, a pattern mirrored by many regulators in other jurisdictions. However, monitoring and measuring insider trading is challenging, and regulators have achieved mixed success in detection and enforcement.[1] Empirically studying such behavior requires systematically identifying situations in which traders may have access to material non-public information and trade in patterns consistent with its use. In our study, we introduce a novel empirical proxy for such situations among large institutional investors, in particular mutual fund managers. Specifically, we examine their endogenous choices to reveal or hide their social ties and relate these choices to their trading performance.

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[1] Assessing insider trading is challenging because only the detected portion is observable and direct evidence is scarce; see Cline and Posylnaya (2019).

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0304-405X/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Focusing on the nuances of Facebook friendships, we find compelling evidence that choosing to hide friendships is strategic rather than random. When fund managers hide their Facebook friends—and these friends include firm officers whose stocks they trade—they realize significantly higher abnormal returns, a pattern further amplified when those firm officers also hide their friends. To our knowledge, we are the first paper to show the significance of the endogenous decision to disclose network information. By choosing to either reveal or obscure their relationships, or by making the availability of such information more or less costly, fund managers and firm officers exercise a subtle yet powerful behavioral choice—one that proves to be more informative than any other publicly observable network feature and, specifically, emerges as the most predictive aspect of networks for future stock prices.

Leveraging information from Facebook (facebook.com), a leading social networking platform with over 200 million monthly U.S. users, this study examines more than 100,000 manually identified profiles of fund managers and firm officers active between 1984 and 2020, along with their 35 million Facebook friends. By analyzing their connections and interactions, we classify each friendship tie as either publicly disclosed or hidden by one or both parties, then analyze how these ties influence investment decisions and subsequent returns. This rich type of information offers several key advantages over traditional proxies for social networks in financial research. Notably, we observe the visibility of each tie and gauge its intensity through user interactions, such as likes, comments, and tags. Our analysis directly captures the extent to which agents in financial markets know and interact with one another, offering a more precise alternative to commonly used indirect proxies—such as geographic proximity or shared employment—which can be imprecise and prone to misclassifying individuals as connected. More importantly, by uncovering hidden connections among financial professionals, we reveal substantial information that would otherwise remain unseen through publicly exposed and observable ties alone. Our findings indicate that these hidden connections hold, on average, the highest value within the network, and overlooking them leads to an incomplete understanding of some of the most pivotal relationships in financial markets.

To illustrate our approach, consider an example from our sample involving Ms. Vinestock, who served as CEO and board member of several firms in the technology sector.[2] One of her friends, Mr. Schiller, is a fund manager of a large actively managed mutual fund, with extensive experience across various buy-side institutions. He describes his investment style as one based on deep fundamental analysis for portfolio selection. Vinestock and Schiller are active Facebook users, but share what we term a *FullyHidden* connection, where both users hide their friends lists, making their connection invisible to conventional social network analysis. They do not share any common network ties via common schools, workplaces, or geographic locations, nor is there documented evidence of a connection. To uncover their connection, we leverage a unique aspect of the Facebook platform: even when users hide their friends lists, a few profile elements remain visible to non-friends but can only be engaged with by friends (see Section 1.3 for details). These engagements can be as subtle as a single like among many on a photo, which may itself be removed or replaced a few months later. By analyzing such traces, we infer friendships between users even when both parties hide their friends lists, leveraging the platform's design that evolved from initially prioritizing connectivity to later incorporating increased privacy features, likely resulting in the observed inconsistencies.

Given their *FullyHidden* connection, we analyze Schiller's performance on trades involving firms where Vinestock held senior roles, including when she became independent chair of a large technology company in mid-2015. At the time, his fund held no position in the stock. Within a quarter (see Fig. 1), it began acquiring shares at around \$20 and continued building its position over the next four years. During this period, the firm experienced several positive developments—including the acquisition of a key rival, European expansion, and new domestic partnerships—that substantially boosted its stock price. Schiller exited the position at over \$200 per share, realizing more than a tenfold return. After his exit, the firm faced significant challenges, including a difficult acquisition integration and new competitors, which drove the stock down by over 60%. Schiller's fund remained out of the stock during this period. Importantly, this was not an isolated case: across different periods and funds, he traded four stocks where Vinestock held management roles, earning an average of 264 basis points per month in abnormal returns (t = 2.31). Zooming out to the broader portfolio level, Schiller's trades linked to hidden connections averaged 208 basis points per month in abnormal returns (t = 2.41), compared to just 12 basis points (t = 0.69) for non-connected holdings—suggesting that his superior performance was unlikely to reflect general systematic ability in portfolio selection.

This example reflects a systematic pattern across the universe of Facebook-identified fund managers and firm officers throughout our full sample period from 1984 to 2020. The most hidden connections—those concealed by both sides of a connected pair—yield abnormal risk-adjusted returns of 135 basis points per month on average (t-stat = 3.54). Yet fund managers hiding their network connections are not inherently better performers: their non-connected holdings generate risk-adjusted returns that are small and statistically indistinguishable from zero. The outperformance of hidden connections also appears uncorrelated with known return determinants, as a long–short strategy's monthly value-weighted alpha of 136 basis points (t-stat = 3.57) is nearly identical to the strategy's raw return of 148 basis points (t-stat = 3.88). Consistent with the impression that hidden connections confer significant informational advantages, we observe abnormal returns moving monotonically with the degree of hiddenness. While connections hidden by both sides generate annualized risk-adjusted returns exceeding 16% (t-stat = 3.54), one-sided concealment is associated with more modest returns: 6.7% per year (t-stat = 1.81) when hidden only by fund managers, and 3.6% per year (t-stat = 1.12) when hidden only by firm officers. In contrast, visibly connected stocks yield risk-adjusted returns of just 1.3% per year, which are statistically insignificant (t-stat = 0.43). Taken together, these findings align closely with the notion of informed trading, suggesting that the degree of connection concealment serves as a proxy for access to material nonpublic information.

The paper also explores the individuals' investment behavior vis-a-vis their connections. In particular, we ask whether fund managers overweight connected stocks and find that the most hidden connections are again those associated with the most abnormal weights. More precisely, while we see a 46% weight increase in stocks of visible connections, the overweight rises to almost 200% for *FullyHidden* connections, and remains statistically large and significant when controlling for time and firm fixed effects—for example, when comparing the weights of two fund managers over the same time period, one of which is visibly connected to the firm's active firm officers, while the other is not.

To explore the mechanism in more depth, we examine the extent to which our results could be driven by either a familiarity or a selection mechanism. For instance, fund managers may prefer to invest in their friends' ventures not because of any information possibly being passed along the network (hidden or not), but because of familiarity toward such stocks. However, a familiarity bias can neither explain the outperformance of connected stocks relative to non-connected stocks, nor why it increases along with the connections' hiddenness.

In contrast, a more refined version of the argument includes selection. Selection might be able to drive both the dispersion in average

[2] The examples are drawn from our sample, but individual names are anonymized.

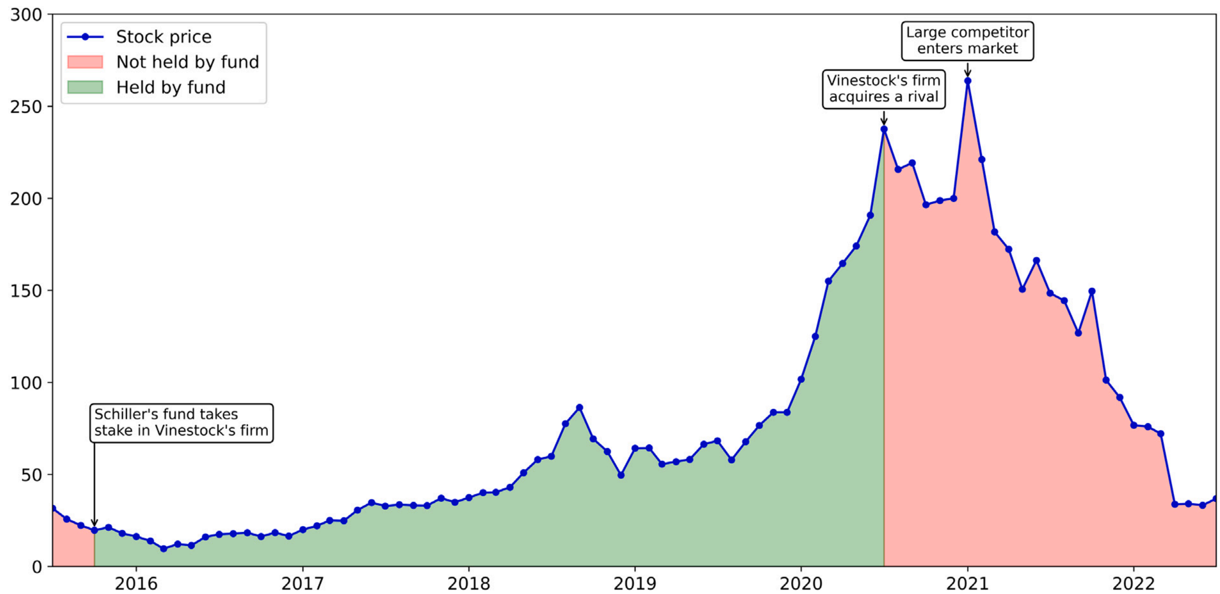

Fig. 1. Holding status of Vinestock's firm's stock by Schiller's fund.
This figure shows the evolution of the stock price of the firm chaired by Vinestock and the holding status of Schiller's fund in this stock between mid-2015 and mid-2022. The blue line with circular markers represents the monthly closing prices of the stock. The green shaded areas indicate periods during which the stock was held by the fund, while the red shaded areas represent periods when the stock was not held by the fund. The y-axis denotes the stock price in USD, and the x-axis represents the time period in months. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

performance between returns earned on connected relative to non-connected stocks and the dispersion in average performance earned on stocks associated with more-hidden relative to less-hidden connections. First, successful individuals may select to or be more likely to jointly match (i.e., the more successful the fund manager, the more likely the firm officer to match with the fund manager, if the firm officer too is successful). Second, unobserved characteristics (e.g., skill or ability) may simultaneously affect both the likelihood of a connection to be hidden and the likelihood of a connection to be associated with performance. In both these scenarios—that is, if our story is either one of a correlation between mutual success and the likelihood of forming friendship ties, or if it is one of an unobserved characteristic which causes hidden connections to select on better quality stock-fund pairs—we would expect to find identical returns on both the hiddenly-connected stocks that the fund managers invest in and the hiddenly-connected stocks that the fund managers choose to avoid (because these two groups sort on the same characteristics). Having said that, we find holdings of such stocks to outperform significantly in times when they are held by fund managers relative to times when they are not held—by 119 basis points in abnormal returns per month (t-stat = 2.59), or 14% per year. This finding is inconsistent with both a selection or even a joint-selection story, and instead strongly supports the notion of hidden connections themselves driving the behaviors we find, with the returns solely being driven by the times of joint holdings between the hiddenly-connected fund manager–firm officer pairs (not a selected characteristic of the two).

Another concern might be that the formation and concealment of these ties could be endogenous to past performance: successful fund managers may attract or curate friendships with successful firm officers ex post. We directly examine this possibility and find no evidence that it drives our results. Facebook-identified fund managers and firm officers are observably similar to their peers, returns accrue only during periods while connected stocks are actually held, and interaction timing points to contemporaneous information flows rather than post-hoc networking.

Additionally, we run a series of supplementary tests and subsample analyses to better understand the mechanisms underlying our results and assess their robustness. While we find that returns are concentrated around corporate news announcements, we also observe that the results are not confined to any specific industry, investment style, or subperiod; instead, they remain large and significant across all of these categories. Moreover, the results are not explicitly concentrated in small stocks—all results we report are value-weighted returns and structurally based on the universe of firms traded by actively managed mutual funds, which biases toward more liquid names. Finally, we examine the persistence of these returns in a multivariate framework that controls for additional return determinants, with coefficients remaining large and statistically significant.

To further probe the mechanism, we show that insider trading cases filed and pursued by the SEC are significantly more likely to involve *FullyHidden* connections. We also decompose the abnormal returns and document that they are highly concentrated around earnings announcements and M&A events—contexts in which access to privileged information is especially valuable. We further show that hidden ties with CFOs are particularly powerful in generating abnormal returns, consistent with CFOs' central role in the shaping and timing of corporate disclosures.

The data observed between January 2020 and August 2023 captures relationships that extend as far back as 1984, thereby reflecting long-standing ties formed decades earlier (e.g., among hometown friends, high school classmates, college roommates, or work colleagues). In this setting, Facebook connections serve merely as footprints of these established ties. However, because Facebook was not introduced until 2004—even though many of the underlying ties predate the platform—one might worry that fund managers appearing earlier in the sample are selectively identified, potentially introducing selection bias. To address this concern, we split the sample into periods before and after 2004. Contrary to the selection hypothesis, we find that the results are actually larger and more statistically significant in the later period, with the effects strengthening steadily through the present day.

Finally, we conduct a placebo test using the same measure of friendship visibility on index funds, where active portfolio decisions are constrained by design, thereby mitigating the potential impact of hidden ties between fund managers and firm officers. We find precisely this to be true: while the effects are strong and robust across actively managed funds, they are absent in index funds. More broadly, we find that the abnormal returns accruing to hiddenly-connected stocks persist for an extended period following the trading. Moreover, and importantly, we observe no sign of any return reversal in the future, indicating that the information associated with these trades is integral to the fundamental firm value and is eventually fully incorporated into it.

Our paper connects to the growing body of empirical literature examining the impact of social ties on information transmission among agents in financial markets. Papers in this spirit include Cohen et al. (2008), who find that fund managers tend to execute larger and more profitable trades on firms managed by personnel with whom they share educational backgrounds; Engelberg et al. (2012), who provide evidence that social connections between firms and banks lead to the issuance of loans with lower interest rates and fewer covenants; Cai and Sevilir (2012), who document that connections between board members of acquirer and target firms are associated with positive announcement returns for the acquirer; Engelberg et al. (2013), who show that CEOs with social connections to outsiders bring more valuable information into the firm and receive higher compensation; Pool et al. (2015), who find that fund managers favor investments in companies that are held by other fund managers living in the neighborhood, amplified by shared ethnicity, Ahern (2017), who provides direct evidence that inside information flows through strong and persistent social ties which underpin illegal insider trading networks; and Jagolinzer et al. (2020), who show that politically connected corporate insiders gain information advantages through government connections.

The remainder of the paper is organized as follows. Section 1 outlines the data sources, sample construction, and summary statistics. Section 2 presents the primary outcomes related to the predictability patterns of returns associated with the hidden friendships identified in our data. Section 3 deepens the analysis by linking the timing and nature of the returns to specific firm news, executive roles, analyst coverage, and SEC enforcement actions. Section 4 conducts robustness tests and explores the horizon of the return effect. Section 5 concludes.

1. Data and sample construction

In this paper, we combine data from multiple sources. To determine the presence and visibility of friendship ties between the individuals in our sample, we use publicly accessible data observed on Facebook. Information on mutual fund holdings, mutual fund returns, and mutual fund managers are obtained from Morningstar Direct (MS Direct). Details on management personnel of stocks held within the fund portfolios are from BoardEx by Management Diagnostics (BoardEx). Stock returns originate from the Center for Research in Security Prices (CRSP). Stock characteristics come from Compustat. Firm-level news data are from RavenPack Analytics (RavenPack). Analyst coverage data are drawn from the Institutional Brokers' Estimate System (IBES).

1.1. Facebook profiles

This study explores how hidden personal relationships between fund managers and firm officers shape mutual fund holdings and returns, using Facebook friendship characteristics as a laboratory network metric. As a first step, we construct an indicator variable that identifies whether a fund manager and a firm officer are connected on Facebook. This process involves locating their Facebook profiles, which are personal pages featuring the user's name, profile picture, cover photo, friends list, timeline, photos, and an "about" section that includes biographical and descriptive details like work history, educational background, places lived, relationship status, and family members. Facebook profiles serve as organizational tools, enabling users to form connections with other users that typically parallel their real-life relationships, such as friends, family, classmates, co-workers, or romantic partners. Users can establish a connection between their profiles by mutually confirming their friendship on the platform. As a result, they will appear in each other's friends list, may have greater access privileges to each other's content, and may receive updates on information produced by or linked with the other.

The process of identifying an individual's Facebook profile can be challenging for a variety of reasons. First, due to Facebook's widespread adoption, many of the attributes that are useful for searching profiles and resolving their creators' identities (e.g., name, workplace, education, and location) are widely shared among Facebook's user base, decreasing their discriminative power. Second, users can restrict the visibility of certain profile attributes by adjusting their privacy settings, which may impede the identification of their profiles and necessitate additional data to support the matching process. Third, users do not necessarily fill in all available data fields, resulting in incomplete profiles with limited identifying information. Lastly, Facebook has imposed limitations on data access in recent times, phasing out broad API availability and confining accessibility to information that is observable through their web interface.[3]

1.2. Matching methodology

As we attempt to identify the Facebook profiles of a large group of individuals, we define a three-step identification procedure to standardize the matching of Facebook profiles. First, for each individual of interest (target) in our sample, we compile a set of potentially matching user profiles (candidates) that exhibit attributes similar to the target's known characteristics. Second, we determine each candidate's probability of matching its respective target by calculating a confidence score based on a variety of similarity and proximity measures. Third, we arrange each target's candidates based on said confidence scores and attempt to manually match the target's true profile from its particular set of candidates. The matching operation was conducted during a single continuous data campaign running from January 2020 to August 2023. We provide a detailed account of this three-step identification procedure in the following description:

In the first step, for each target in our sample, we assemble a list of candidates that exhibit attributes consistent with the target's known characteristics. These attributes may include, for example, location, workplace, or education.

In the second step, for each candidate that is associated with a target, we calculate a confidence score indicating the likelihood of the identity behind the candidate being equivalent to the target. The score is calculated based on a range of measures representing similarities between the candidate and the target. With each measure, we focus on capturing a different aspect of potential similarity. Using semantic measures, we analyze a candidate's various profile attributes (e.g., screen name,[4] username, education, workplace, location) and compare their values to those held by the target. Before the comparison of attributes, we selectively augment the target's attributes with their semantically equivalent representations, if applicable. For instance, the name value "Robert" may be augmented by "Rob" and "Bob", the alma mater value "University of Mississippi" may be augmented by "Ole Miss", and the employer value "Alphabet" may be augmented by "Google". For some measures, in addition to looking for perfect matches between

[3] Major endpoints of Facebook's main API ("Graph API") were deprecated in April 2018. Further restrictions were imposed in June 2019 when the semantic search engine ("Graph Search") was disabled. These measures have limited the capacity to read Facebook's social graph and associated information.

[4] Facebook's real-name policy mandates users to use the name that they are commonly known by in their everyday life. The name must correspond with their identification documents, which they must submit upon request to verify their identity. If a nickname is used, it must be a derivative of the user's authentic name.

entire strings of attributes, we consider flexible matching schemes to capture partial overlaps between the attributes' meaningful units. For various other measures, we use attributes that are not observed but inferred from information associated with the user. For example, for candidates with a Facebook user ID between zero and 350 million, we leverage our finding that these IDs were not assigned sequentially but instead segmented by college, reflecting the period when Facebook membership was restricted to individuals with email addresses from selected schools. Based on this structure, we may infer the educational institution of the candidate from the numeric range of their user ID, irrespective of whether the education attribute is visible on the profile.[5]

In addition to inferring attributes from the user's information, we may also draw inferences from information about their connections. Specifically, for certain measures, we examine the plurality of the user's friends and determine the most frequently appearing attribute value among them. For example, if a significant percentage of a candidate's friends attended a certain college or reside in a certain city, we infer that the user is likely associated with that institution or location. Each inferred attribute is weighted by a confidence reflecting its reliability. If, for instance, the alma mater is inferred based on a large number of the candidate's friends having attended this institution, the confidence in the inferred attribute is high; otherwise, it is low. Some measures are adjusted to reflect the rarity of a match. For example, if the alma mater "Coe College" of a candidate with the common name "James Miller" matches the alma mater of the target, we denominate the probability of the match by the number of Coe College graduates in our sample by the name of James Miller. Following the calculation of the above measures, we aggregate the values produced for each candidate–target pairing into a single confidence score.

In the third step, we identify a target's genuine profile from the pool of scored candidates, where applicable. In order to conserve human resources, any candidate whose confidence score does not exceed a predetermined threshold is eliminated. The remaining candidates for a particular target are ranked based on their respective scores and reviewed in descending order, beginning with the highest-ranked candidate and progressing to lower ranks. To ensure accuracy, visual confirmation is generally required to establish a match.[6] If a candidate's privacy settings restrict such confirmation, we attempt to validate the match indirectly by linking the candidate to an immediate family member of the target (i.e., spouse, parent, or child). If we identify the Facebook profile of such a family member and determine—based on an observable friends list—that the target is not among their friends, we conclude that the target does not maintain a Facebook profile, and no further matching is conducted.

1.3. Friendship ties

After identifying the Facebook profiles of our sampled individuals, two complementary approaches are adopted to determine their friendship connections.

The first approach involves identifying all users listed in the individuals' friends lists. If a particular user's friends list is not publicly observable, we may still be able to determine some of their connections through "backlinks" from their friends' friends lists.[7] The disclosure of backlinks may be enhanced by drawing on Facebook's Mutual–Friends feature, which requires two user IDs A and B as input and returns a list of mutual friends if certain conditions are met. Specifically, if A's friends list is private while B's friends list is public, the feature can return all of B's friends who are connected with A and who publicly disclose the connection on their end. A recursive iteration can then be implemented, pairing each target with users likely to share mutual friends with the target (pivot users), and using all newly identified friends returned at each step as additional pivot users for further iterations. Once initiated for a given target, such an iteration would continue until all pivot users are paired with the target and no further friends are disclosed. The feature itself was deactivated by Facebook in August 2021.

The second approach infers friendship ties from user interactions. On the platform, although users may choose to hide their friends list, this does not entirely remove all traces that their friends may leave on publicly accessible parts of their profile, which can still make the friendship identifiable. In particular, Facebook users can interact with each other through likes, comments, and tags on posted content, and the posting user may restrict the audience that can see the content, engage with it, or see who has engaged with it (e.g., limiting its visibility to their friends). While such content is then no longer visible to the general public, the visibility of a small subset of profile items classified as "public information" cannot be restricted by the user, including the currently selected profile picture and cover photo. These images remain visible even to non-friends for as long as they are selected; however, the set of users who can interact with them is still, by default, limited to the user's friends. This unique constellation enables us to infer connections between users by observing which accounts react to these images, even if both sides hide their friends lists, provided that certain additional criteria are met. For instance, a photo cannot be used for evaluation if other users besides the profile holder are tagged in it, as the audience of this photo (i.e., those who can interact with it) will automatically extend to include the friends of the tagged users. Friendships derived from such interactions may be visible only for a limited period of time, until the particular photo is replaced. It will then no longer fall into the public information category and thus may no longer be visible to non-friends.

After determining the friendship ties between our sample individuals, we proceed by classifying the degree of visibility of these connections, which produces the key variable for our study (*FriendshipVisibility*). This variable is a vector of four dummy variables capturing the degree of visibility of a Facebook friendship observed between a fund manager and a firm officer depending on whether the friendship is publicly observable through both the fund manager's and the firm officer's friends lists (*FullyVisible*); observable through the fund manager's friends list but not through the firm officer's friends list (*FirmOfcrHides*); not observable through the fund manager's friends list but observable through the backlink from the firm officer's friends list (*FundMgrHides*); or neither observable through the fund manager's friends list nor through the firm officer's friends list (*FullyHidden*).

1.4. Mutual fund sample

In this section, we detail the construction of the sample of funds and fund holdings for our study. The universe of fund-month observations whose fund managers' Facebook profiles we identify will determine the relevant stock-month observations, which in turn will span the universe of firm officers whose Facebook profiles we are interested in matching.

[5] Each Facebook user is assigned a unique numeric ID upon registration. These IDs were assigned based on the registration email domain of the educational institution, with distinct ID ranges allocated to specific schools. For example, registrants affiliated with the University of Waterloo (students and alumni using "@uwaterloo.ca") were assigned IDs between 122,600,000 and 122,699,999. We manually identify such clusters for 2,708 schools in the Facebook user ID space between zero and 350 million.

[6] The validation of Facebook profiles was aided by trained research assistants.

[7] Facebook is structured as an undirected graph in which mutual consent is required for a connection to form between two users. Thus, a connection between two users A and B can be disclosed with certainty either by showing that A is connected to B or by showing that B is connected to A.

The original sample of funds contains the universe of actively managed U.S.-domiciled U.S. equity mutual funds covered by MS Direct. While most prior studies in the mutual fund literature rely on the Thomson Reuters Mutual Fund Holdings (TR Holdings) database for holdings data, we chose Morningstar for several reasons. First, we find that fund holdings data from MS Direct are more frequently available than those from TR Holdings. Second, we corroborate prior claims that MS Direct provides a more comprehensive representation of the actual compositions of the funds' portfolios (e.g., Elton et al. (2011)). Third, MS Direct are recognized for their higher accuracy in reporting the funds' portfolio managers (e.g., Massa et al. (2010) and Patel and Sarkissian (2017)).

We construct our sample of funds by including both defunct and active fund share classes to avoid a potential survivorship bias. We obtain the funds' historical Morningstar categories from historical time series data. To ensure an equitable comparison basis for fund managers, we limit the sample to domestic and actively managed U.S. equity funds (i.e., we exclude index funds, international funds, money market funds, and funds that focus on bonds, commodities, nontraditional equity, or alternative asset classes). We adhere to standard protocol by eliminating funds with titles that include the terms "index" or "idx". Since funds may be offered in various share classes that differ in their fee structures but provide exposure to the same portfolio of securities, we consolidate observations for funds with multiple share classes into one. For each fund that passes the aforementioned filters, we pull historical management data from MS Direct, which includes the names of the fund managers, start and end dates of their management periods, brief biographies, and educational background information. We exclude fund-month observations for which these data are unavailable. For the stocks held by the funds, we obtain one-month-ahead return data from the CRSP Monthly Stock Files and link these returns to the funds' holdings using historical CUSIP numbers. Following this procedure, our sample comprises 415,676 fund-month observations spanning January 1984 through December 2020. This sample includes 5,053 unique funds, with an average of 1,386 funds per calendar quarter. We use this sample, hereafter referred to as the benchmark universe of funds, when constructing weights for our benchmark portfolios.

1.5. Fund manager sample

The 5,053 mutual funds that pass the initial filtering process are managed by a total of 10,036 fund managers.[8] To prepare for matching their Facebook profiles according to the methodology outlined in Section 1.2, we collect additional biographical data on the individuals. We begin by determining their most complete names (including middle names, nicknames, birth names, surnames adopted upon marriage, and suffixes) using the Financial Industry Regulatory Authority's BrokerCheck database, the Securities and Exchange Commission's (SEC's) Investment Adviser Public Disclosure database, and the Chartered Financial Analyst's (CFA's) Institute's member directory. Next, we gather information on vital dates, portraits, places of residence, family members, academic degrees, graduation years, and occupational backgrounds through a cross-database search of various sources such as LinkedIn profiles, Bloomberg executive profiles, The Wall Street Transcript profiles, biographies published by fund firms, filings with the SEC, obituaries on legacy.com, alumni publications on ancestry.com, and newspaper articles on newspapers.com.

We then proceed with matching the fund managers' profiles on Facebook and successfully identify 4,198 (41.8%) out of the 10,036 fund managers present in the benchmark universe of funds. This ratio aligns well with general figures on Facebook usage in the United States, which suggest that approximately six in ten adults use or have used Facebook at some point in their lives. The left subfigure of Fig. 2 illustrates the percentage of Facebook-identified fund managers with respect to the total number of fund managers meeting the pre-established criteria in the benchmark universe throughout the sample period. Note that the upward trend in this percentage over time reflects the growing share of fund managers who entered the sample in later years—particularly the 2000s onward—and who are more likely to have active Facebook profiles. We then restrict the sample of funds to only those managed by fund managers whose Facebook profiles we are able to identify. Team-managed funds are included if we identify the Facebook profile of at least one team member. These steps result in a sample reduction to 265,320 of 415,676 fund-month observations and 4,100 of 5,053 funds.

1.6. Firm officer sample

We obtain employment data and biographical information on the firm officers leading the firms whose stocks are held by our sample of Facebook-identified funds from BoardEx, a data purveyor that collects and consolidates public domain information on management personnel of publicly quoted and large private companies in North America and around the world. BoardEx data come from multiple sources, such as the SEC, press releases, primary websites, and U.S. stock exchanges, and have been used in various economic studies to analyze the impact of social networks (Cohen et al., 2008, 2010; Engelberg et al., 2012; Chen et al., 2020). BoardEx offers comprehensive overviews of the composition of boards and senior management, with data collecting beginning in 1999, although most individual company records extend back further in time. BoardEx provides information on the current and past roles of firm officers at active and inactive firms, including start/end dates for their roles, educational backgrounds, and affiliations with charitable or volunteer organizations.

BoardEx data are categorized into four geographical regions to which firms are allocated based on the most recent location of their headquarters.[9] To assemble the sample of firm officers for this study, we begin by combining the management personnel of both currently and formerly U.S.-headquartered public companies from the four files into one file. We proceed by merging the data with the funds' portfolio holdings using historical linking tables furnished by Wharton Research Data Services (WRDS), which provide a link between the respective firm identifiers of BoardEx (companyid) and CRSP (permco). We discard employment records for which BoardEx does not provide the start date of an individual's role at a company. If BoardEx does not specify an end date for a role, we adhere to the provider's instructions and assume that the position is still held by the individual. Additionally, we exclude individuals for whom BoardEx's records indicate that they passed away prior to Facebook's launch in February 2004. After applying the aforementioned filters, we are left with 281,829 distinct firm officers involved in the management of 8,646 distinct firms held by the Facebook-identified funds. Next, for the purpose of matching the

[8] Our fund manager sample is subject to two adjustments. First, we address instances where multiple individual records pertain to the same entity by consolidating them (309 individuals were consolidated into 153). These duplications often arise due to name changes (e.g., Katherine Lieberman née Buck being referred to as either "Katherine Buck" or "Katherine Lieberman") or pseudonyms (e.g., Langton C. "Tony" Garvin being referred to as either "Langton C. Garvin" or "Tony Garvin"). Second, we exclude any fund managers who passed away before the launch of Facebook in February 2004 (39 individuals). Those who died after the threshold are retained in the sample, as they would have had the opportunity to create a Facebook profile that may still be accessible or preserved online.

[9] BoardEx assigns each firm to one of four geographically-themed files—North America, United Kingdom, Europe, Rest of World—based on the current location of its headquarter. For instance, if a firm's headquarter is relocated from the U.S. to Europe, BoardEx will transfer all observations associated with that firm into the Europe file, including those pertaining to years when the firm was headquartered in the U.S.

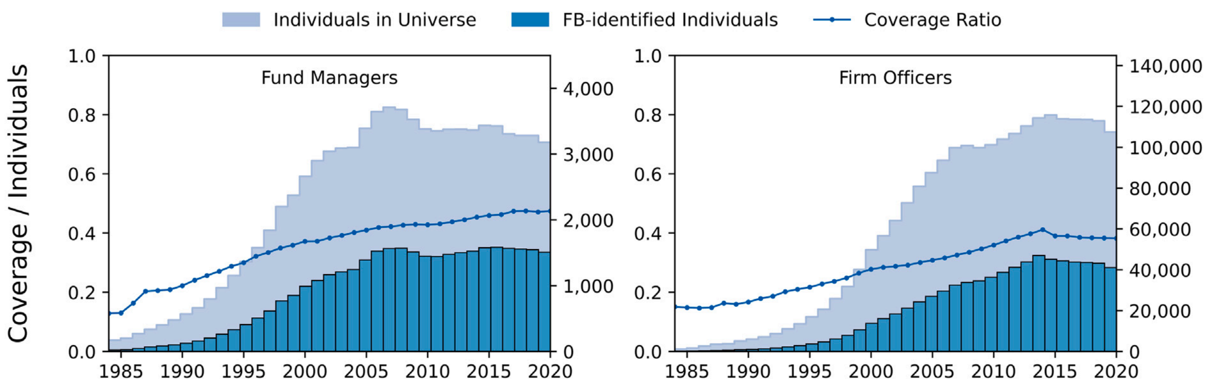

Fig. 2. Ratios of Facebook-identified fund managers and firm officers.
This figure presents annual coverage rates of Facebook-identified fund managers and firm officers over the 1984–2020 sample period. Panel A shows identification coverage for fund managers, based on individuals populating the benchmark universe of mutual funds defined in Section 1.4. Panel B shows identification coverage for firm officers, based on the benchmark universe of firms defined in Section 1.6. In each panel, the left *y*-axis indicates the proportion of individuals identified on Facebook (ranging from 0 to 1), while the right *y*-axis shows the number of individuals per year. The stacked bars represent the annual counts of individuals in the respective benchmark universe (light bars) and those identified on Facebook (dark bars). The solid line traces the corresponding identification ratio—the share of Facebook-identified individuals each year. Across the full sample period, 4,198 of 10,036 fund managers and 101,866 of 281,829 firm officers were identified on Facebook.

firm officers' profiles on Facebook, we collect their biographical information from various BoardEx files including comprehensive names, educational backgrounds and work histories.

With this data available, we successfully identify the Facebook profiles of 101,866 (36.1%) of the 281,829 firm officers in the full sample. These individuals are managing 8,444 of the 8,646 firms in the full sample. The right subfigure of Fig. 2 illustrates the percentage of Facebook-identified firm officers with respect to the total number of firm officers meeting the pre-established criteria in the benchmark universe throughout the sample period.

1.7. Descriptive statistics

This study leverages a uniquely large body of individual-level information observed on Facebook. Between January 2020 and August 2023, we identified 4,198 fund managers and 101,866 firm officers along with their profile information. Many of the underlying friendships were formed decades earlier—some tracing back to high-school cohorts in the early 1980s—but the Facebook variables we analyze reflect each profile as it appeared during the observation window, thereby providing the richest available cross-section of contemporaneous visibility settings and interactions.

Across these profiles, a wide range of attributes is observable. Where available, these include screen names, usernames (unique alphanumeric identifiers), user IDs (unique numerical strings), gender, date of birth, contact information, relationship status, number of friends, the screen names and usernames of friends, family members, and relationships, details on workplaces, high schools, and colleges attended, information on current city, hometown, and places lived, details on life events, and all uploaded photos, including profile pictures, cover photos, photos in miscellaneous albums, and timeline posts with photographic elements. Each photo is examined for upload date, user-provided descriptions, and the screen names and usernames of any users who are tagged in it, expressed liking toward it, or provided comments on it, as well as the contents of said comments.

In Table 1, we present a summary of profile-level data available on the Facebook profiles of fund managers and firm officers. For each data category, we report both the percentage share and the total number of profiles in which such information was observed. The profiles of the two cohorts exhibit no substantial differences in terms of the information displayed. Screen names, usernames, and user IDs are fully available across the sample, as these attributes cannot be concealed from the public. Friends lists are accessible for slightly more than 50% of both fund managers' and firm officers' profiles. Profile pictures and cover photos are observable in about 90% of profiles, respectively, while additional photos, such as those uploaded to miscellaneous albums, are displayed in about 70% of profiles. The data types typically provided by users to assist their real-life friends in finding and connecting with them on the platform, such as work, college, current city, and hometown, are prevalent in the mid-thirties to lower fifties range. The availability of more sensitive data types, such as relationship status and family members, is less frequent, with percentages ranging in the lower twenties. It is noteworthy that the quality of the disclosed information may vary widely, depending on various factors such as the method by which the user submitted the data.[10]

In Table 2, we present descriptive statistics on item-level data associated with the sample individuals' Facebook friends, photos, and reactions. For each variable, we report the mean, median, standard deviation, total number of data items observed, and the number of profiles in which such items appeared within that category. The statistics are computed based on observations with non-missing values. Across the sample, we count a total of 35.2 million Facebook friends for our sample individuals, with 1.1 million associated with 4,066 of the 4,198 fund managers and 34.1 million associated with 100,193 of the 101,866 firm officers. The median number of friends per fund manager and firm officer is 167 and 188, respectively. These friends stem not only from the individuals' profiles but also from others sources, such as backlinks of other users' friends lists and reactions given to photos uploaded by other users on their profiles (refer to Section 1.3 for our approaches to assessing friends). We report figures on friends segmented by the different approaches through which they were observable, including friends observed in publicly disclosed friends lists (*Friends–Published*), friends inferred through backlinks from other users' friends lists (*Friends–Backlinks*),[11] and friends inferred from reactions

[10] By way of example, the user may have entered certain information manually ("Went to Bayreuth"), or declared it through a menu interface designed to assist users in aligning an entry with a concept that is known to Facebook ("Studied at Uni Bayreuth"). In the latter case, a standardized entry is established, which includes a hyperlink directing to the Facebook page devoted to the corresponding concept.

[11] The lower incidence of disclosed friends for firm officers via backlinks arises from Facebook's discontinuation of the Mutual-Friends feature in August 2021 (see Section 1.3). However, because connections for all fund managers had already been established while the feature was active, the connection counts between fund managers and firm officers in this study remain unaffected.

Table 1
Profile-level Facebook data.

	Fund Managers (N Profiles = 4,198)		Firm Officers (N Profiles = 101,866)	
	% Share	N Profiles	% Share	N Profiles
Screen Name	1	4,198	1	101,866
Username	1	4,198	1	101,866
User ID	1	4,198	1	101,866
Gender	.79	3,320	.71	72,127
Date of Birth	.01	37	.02	1,884
Friends List	.56	2,353	.53	54,423
Relationship Status	.21	885	.21	21,542
Family Members	.24	988	.23	23,554
Profile Picture	.90	3,793	.94	96,153
Cover Photo	.88	3,707	.89	90,864
Other Photos	.67	2,815	.77	78,023
Work	.33	1,379	.36	36,774
College	.46	1,939	.52	52,776
High School	.37	1,548	.44	45,070
Current City	.53	2,245	.48	49,172
Hometown	.43	1,815	.43	43,428
Other Places Lived	.22	919	.17	17,818
Life Events	.25	1,054	.33	33,209

This table presents the full set of profile-level data fields observable on the Facebook profiles of the 4,198 fund managers and 101,866 firm officers included in our sample, which covers 4,100 mutual funds and their stock holdings over the period 1984–2020. These data fields include basic identifiers (screen name, username, user ID), demographic attributes (gender, date of birth), social network indicators (friends list, relationship status, family members), visual content (profile picture, cover photo, other photos), educational and professional history (work, college, high school), and geographic background (current city, hometown, other places lived), as well as life events. For each data field, we report the percentage of profiles in which the field is populated (% Share) and the number of such profiles (N Profiles), separately for fund managers and firm officers.

Table 2
Item-level Facebook data.

	Fund managers (N Profiles = 4,198)					Firm officers (N Profiles = 101,866)				
	Mean	Median	SD	N Items	N Profiles	Mean	Median	SD	N Items	N Profiles
Friends	269	167	394	1,095,018	4,066	341	188	635	34,148,303	100,193
Friends–Published	310	213	425	728,897	2,353	443	275	561	24,102,607	54,423
Friends–Backlinks	110	66	144	180,951	1,648	10	5	19	402,652	41,613
Friends–Reactions	126	69	199	180,058	1,425	196	111	664	7,932,927	40,540
Photos	56	6	321	187,328	3,374	110	12	420	9,831,174	89,294
Reactions	663	118	2,601	2,138,914	3,226	695	215	2,039	59,448,651	85,512
Likes	525	91	2,094	1,666,017	3,173	598	184	1,752	50,593,776	84,627
Comments	148	33	537	435,814	2,954	105	35	322	8,283,007	78,817
Tags	37	5	144	37,083	1,014	17	5	67	571,868	33,803

This table reports descriptive statistics on item-level Facebook data observed for friends, photos, and reactions associated with the 4,198 fund managers and 101,866 firm officers in the Facebook-identified sample of 4,100 mutual funds and their stock holdings in the period 1984–2020. The statistics reported for each variable include the mean, median, standard deviation (SD), total number of data items observed (N Items), and number of profiles in which such items were observable within that category (N Profiles). Statistics are computed based on observations with nonmissing values. Figures reported on friends are further segmented by the different approaches through which they were observed (refer to Section 1.3 for our approaches to observing friends), including friends identified through publicly disclosed friends lists (Friends–Published), friends identified through backlinks from other users' friends lists (Friends–Backlinks), and friends inferred from reactions received on or given to restricted photos (Friends–Reactions). Photos include profile pictures, cover photos, and those in miscellaneous albums. Reactions refer to engagements with photos uploaded to profiles, excluding the reactions users gave to their own uploads.

received on or given to restricted photos (*Friends–Reactions*). Additionally, we report statistics on observable photos which include a total of ten million photos uploaded by the sample individuals. Of these, 187 thousand photos appeared on 3,374 of the 4,198 fund manager profiles, and 9.8 million photos appeared on 89,294 of the 101,866 firm officer profiles. Lastly, the table presents statistics on reactions (i.e., likes, comments, and tags) received by these uploaded photos, excluding reactions that the individuals gave themselves to their uploads. In total, we observe 61.6 million reactions across the sample individuals, with 3,226 of the 4,198 fund managers receiving 2.1 million reactions, and 85,512 of the 101,866 firm officers receiving 59.4 million reactions. Of these reactions, 85% are likes, 14% are comments, and 1% are tags.

In Table 3, we provide summary statistics reflecting the annual composition of the Facebook-identified sample of funds, including details on fund managers, the funds' common stock holdings, and management personnel of the firms whose stock is held by the funds. The sample comprises 265,320 fund-month observations managed by 4,198 Facebook-identified fund managers in the period 1984–2020. The Morningstar benchmark universe used to calculate percentage coverages for the funds is the sample consisting of 415,676 fund-month observations whose construction we detail in Section 1.4. The BoardEx benchmark universe used to calculate percentage coverages for firm officers is the sample constructed in Section 1.6. The statistics reported for each variable include the mean, median, minimum, maximum, and standard deviation. In median terms, the sample includes 1,508 funds per year, accounting for an annual coverage of 59% of the 5,053 funds in the benchmark universe over the study period or 53% of the benchmark universe's total net assets under management. The sample of Facebook-identified fund managers includes 1,166 individuals per year, corresponding to an annual coverage of 38% of the 10,036 fund managers active in the benchmark universe during the study period. The Facebook-identified funds invest in 4,267 firms per year,

Table 3
Summary statistics for Facebook-identified funds.

	Mean	Median	Min.	Max.	SD
Facebook-identified Funds (FB Funds)	1,126	1,508	23	1,985	755
% of Funds in MS Universe	.53	.59	.15	.75	.19
% of TNA in MS Universe	.50	.53	.09	.79	.22
Facebook-identified Fund Managers	934	1,166	22	1,584	614
% of Fund Managers in MS Universe	.35	.38	.13	.47	.11
Stocks held by FB funds	3,768	4,267	310	5,314	1,443
% of Stocks in CRSP Universe	.50	.57	.05	.70	.19
% of Market Cap. in CRSP Universe	.85	.96	.32	1	.18
Firm Officers held by FB Funds	59,949	64,050	1,122	115,811	45,273
% of Firm Officers in BoardEx Universe	.95	.99	.56	1	.10
Facebook-identified Firm Officers	22,116	18,405	169	55,596	19,903
% of Firm Officers held by FB Funds	.29	.29	.15	.52	.11

This table provides summary statistics reflecting the annual composition of the Facebook-identified sample of mutual funds (FB Funds), fund managers, the funds' common stock holdings, and management personnel of the firms whose stock is held by the funds. The sample comprises the 265,320 fund-month observations managed by the 4,198 Facebook-identified fund managers in the period 1984–2020. The statistics reported for each variable include the mean, median, minimum, maximum, and standard deviation (SD). The Morningstar benchmark universe (MS Universe) used to calculate percentage coverages for the funds is the sample whose construction we detail in Section 1.4. The BoardEx benchmark universe used to calculate percentage coverages for firm officers is the sample constructed in Section 1.6.

representing an annual coverage of 57% of the stocks covered by the CRSP universe and covering 96% of the CRSP universe's market capitalization. These firms are managed by 64,050 firm officers per year, or an annual coverage of 99% of the 281,829 firm officers present in the BoardEx benchmark universe. The sample of Facebook-identified firm officers includes 18,405 individuals per year, accounting for an annual coverage of 29% of the aforementioned firm officers whose firms' stock is held by the Facebook-identified funds.

In Table 4, we present data on the Facebook friendships that we observe between the fund managers and firm officers in our sample, broken down by the friendships' visibility levels as defined in Section 1.3. We report on three types of fund manager–firm officer friendships: total friendships (*Friendships–All*), tradable friendships (*Friendships–Tradable*), and traded friendships (*Friendships–Traded*). A friendship is deemed "tradable" if the fund manager's tenure at the fund coincides with the firm officer's tenure at the firm and if the firm's stock, during the given month, is held by at least one fund in the same Morningstar category as the relevant fund. Similarly, a friendship is deemed "traded" if the fund manager's fund holds the firm's stock during the firm officer's tenure at the firm. We provide information on tradable friendships (in addition to total and traded friendships) as supplementary information to enhance the context of the results. Tradable friendships exclude those that could not have been traded due to the connected fund manager and firm officer not simultaneously managing the fund and the firm, as well as friendships that are unlikely to be traded as per the fund's objectives (approximated through holdings of peers in the same Morningstar category), such as the friendship between a fund manager of a large-cap value fund and a firm officer of a small-cap growth firm. The table reports figures on the number of connected pairs, the number of distinct fund managers and firm officers involved in these pairs, the number of stocks associated with the pairs, and the number of underlying fund-month-stock observations. In total, our analysis reveals 18,478 Facebook friendships between 2,803 fund managers and 10,748 firm officers. With regard to tradable friendships, we observe 9,012 friendships between 2,107 fund managers and 6,050 firm officers. Of these friendships, 2,908 (32%) are traded, involving 1,052 fund managers and 2,143 firm officers. When categorizing the traded friendships based on their visibility levels, 930 (32%) are assigned to the *FullyVisible* category, 959 (33%) are classified as *FundMgrHides*, 758 (26%) are classified as *FirmOfcrHides*, and 261 (9%) are classified as *FullyHidden*. Notably, a comparison between the tradable and traded friendships indicates that the desire to hide Facebook friends lists coincides with a higher likelihood of these friendships being activated. Specifically, while only about 28% of the *FullyVisible* friendships are traded, this percentage increases to 31% for *FirmOfcrHides* friendships and to 37% and 44% for *FundMgrHides* and *FullyHidden* friendships, respectively.

Table 4 also suggests that the sample of connected pairs is not dominated by a few super-connectors but instead involves a large number of individuals from both groups—fund managers and firm officers. These individuals are dispersed across various funds and firms. To illustrate this distribution, Fig. 3 presents a network graph depicting Facebook-identified connections between fund managers (represented by blue nodes) and firm officers (represented by red nodes) classified as tradable at any point throughout our sample period. The darker shades within the graph indicate traded connections within the subset of tradable connections. Each node represents an individual, and edges signify Facebook friendships between fund managers and firm officers. Individuals are clustered by their current or most recent employer. In cases of multiple affiliations, individuals are assigned to the firm where they hold their most senior role. This graph visually demonstrates the extensive and intricate network of connections, emphasizing the dispersion and complexity of these relationships across different funds and firms. It underscores the widespread nature and potential influence of these hidden connections on fund managers' investment decisions.

2. Main results

2.1. Portfolio weights

In this section, we examine whether fund managers' hidden friendships have an effect on the funds' holdings. If fund managers with non-public connections to company executives are better informed, they may allocate a larger proportion of their funds to securities managed by those individuals. To assess this possibility, we calculate the portfolio weights in Facebook-connected stocks for each fund-month observation as the dollar investment in these stocks divided by the fund's total dollar holdings during that period. We then estimate various forms of the regression equation

$$w_{i,k,t} = \alpha_0 + \beta' FriendshipVisibility_{i,k,t} + \Gamma' Controls_{i,k,t} + \epsilon_{i,k,t}, \quad (1)$$

where $w_{i,k,t}$ is the weight of fund i in stock k at time t; $FriendshipVisibility_{i,k,t}$ is a set of dummy variables reflecting the visibility of the Facebook friendship between a fund manager of fund i and a firm officer of firm k, as defined in Section 1.3; and $\Gamma' Controls_{i,k,t}$ is a vector of control variables including *Style*, the percentage of the fund's total net assets invested in the style corresponding to the stock being considered (style is calculated as in Daniel et al. (1997)), market value of equity (*ME*), book to market (*BM*), and past 12-month return

Table 4
Statistics on Facebook friendships by visibility.

	N Pairs	N Fund Mgrs.	N Firm Offcs.	N Stocks	N Obs.
Friendships–All	18,478	2,803	10,748	–	–
FullyVisible	6,748	1,583	4,590	–	–
FirmOfcrHides	6,211	1,438	3,968	–	–
FundMgrHides	4,235	927	3,008	–	–
FullyHidden	1,284	426	993	–	–
Friendships–Tradable	9,012	2,107	6,050	3,371	–
FullyVisible	3,298	1,105	2,516	2,043	–
FirmOfcrHides	3,084	1,007	2,253	1,730	–
FundMgrHides	2,038	658	1,600	1,354	–
FullyHidden	592	273	486	461	–
Friendships–Traded	2,908	1,052	2,143	1,510	171,755
FullyVisible	930	459	768	755	45,542
FirmOfcrHides	959	441	779	680	45,020
FundMgrHides	758	343	618	576	51,350
FullyHidden	261	136	214	245	29,495

This table presents data on the Facebook friendships that we observe between the fund managers and firm officers in our sample, broken down by the friendships' visibility levels. We report on three types of fund manager–firm officer friendships: total friendships (*Friendships–All*), tradable friendships (*Friendships–Tradable*), and traded friendships (*Friendships–Traded*). The table reports figures on the number of connected pairs (*N Pairs*), the number of distinct fund managers (*N Fund Mgrs.*) and firm officers (*N Firm Offcs.*) involved in these pairs, the number of stocks associated with the pairs (*N Stocks*), and the number of underlying fund-month-stock observations (*N Obs.*). A friendship is deemed "tradable" if the fund manager's tenure at the fund coincides with the firm officer's tenure at the firm and if the firm's stock, during the given month, is held by at least one fund in the same Morningstar category as the relevant fund. A friendship is deemed "traded" if the fund manager's fund holds the firm's stock during the firm officer's tenure at the firm. We denote the visibility of a Facebook friendship depending on whether the friendship is publicly observable through both the fund manager's and the firm officer's friends lists (*FullyVisible*); observable through the fund manager's friends list but not through the firm officer's friends list (*FirmOfcrHides*); not observable through the fund manager's friends list but observable through the backlink from the firm officer's friends list (*FundMgrHides*); or neither observable through the fund manager's friends list nor through the firm officer's friends list (*FullyHidden*).

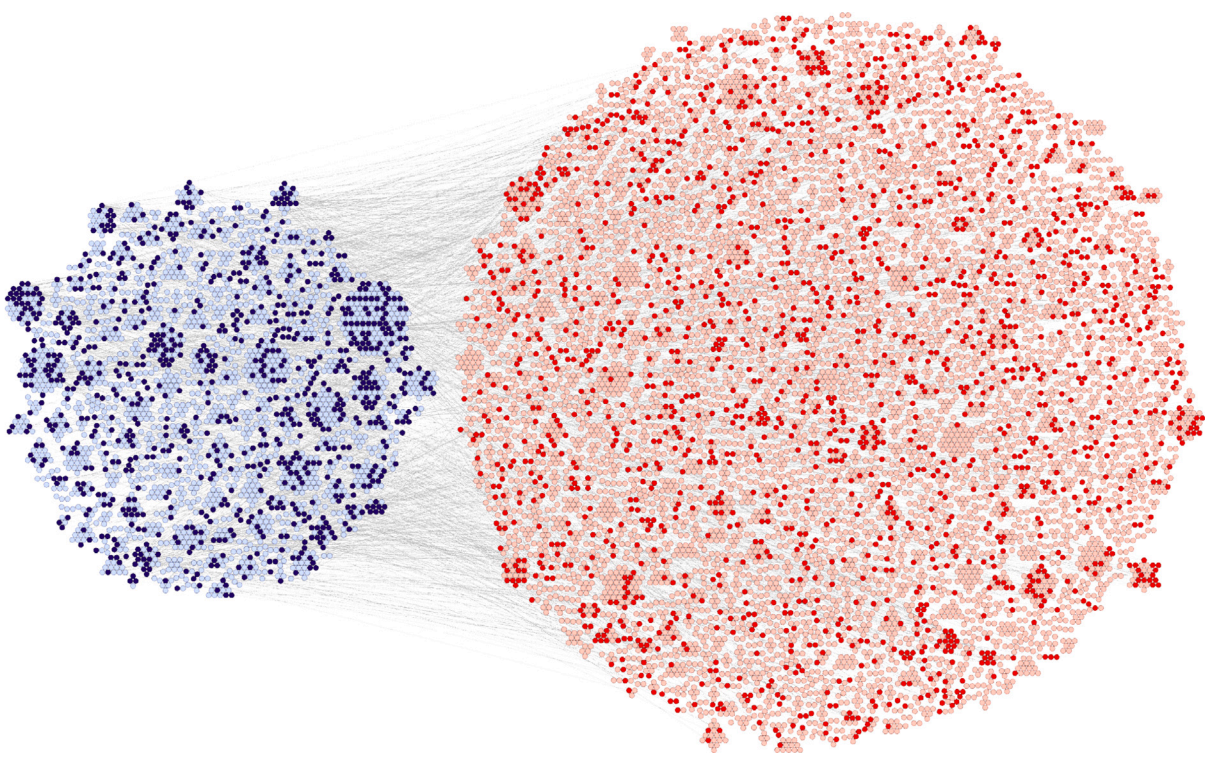

Fig. 3. Network of tradable friendships between fund managers and firm officers.
This figure visualizes Facebook-identified friendships between fund managers and firm officers that were tradable at least once during the 1984–2020 sample period (see Section 1.7 for the tradability definition). Blue nodes represent fund managers; red nodes represent firm officers. The graph is static and pools all such relationships throughout the entire sample period. The darker shades represent fund manager-firm officer ties with traded connections within the subset of tradable connections. Each node is labeled by its current (or most recent) employer and is assigned to that organization's cluster; in cases of multiple affiliations, we use the role in which the individual holds the greatest seniority. Edges denote Facebook friendships; their lengths have no economic interpretation. The layout is produced with a circle-packing algorithm that groups nodes by employer and then arranges the clusters within the overall graph. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(*R12*). If fund managers tilt their portfolios toward the firms managed by their non-public firm officer friends, we would expect to find that β is positive and statistically significant.[12]

In Table 5, we present coefficient estimates and standard errors clustered at the fund and period levels from Panel OLS estimations of various specifications of Eq. (1). All regressions include period fixed effects, and the unit of observation is stock-fund-month. The main results are shown in columns 1–4, each including one visibility indicator from the set $FriendshipVisibility_{i,k,t}$ and a constant in the regression. Columns 1–4 reveal over-allocations to securities of friends that vary significantly depending on whether or not the friendship between a fund manager and a firm officer is publicly visible through their friends lists. Specifically, fund managers invest an additional 35.2 basis points in securities of firms managed by firm officers friends in the *FullyVisible* category, compared to the average weight of 74.6 basis points. They allocate 63.9 additional basis points to securities of friends in the *FirmOfcrHides* category, 95.8 additional basis points to securities of friends in the *FundMgrHides* category, and 136.5 additional basis points to securities of friends in the *FullyHidden* category. In columns 5 and 6, the regressions from columns 1 and 4 are estimated with fund fixed effects, focusing solely on variation at the stock level (i.e., firm officer changes). While fund fixed effects explain the variation in fund managers' portfolio allocations toward *FullyVisible* friends, the coefficient on *FullyHidden* friends remains statistically significant at 50.9 basis points. Finally, columns 7 and 8 estimate both specifications with firm fixed effects to control for the average weight funds have in each stock. This approach relies on the variation at the fund level over time (i.e., fund manager changes) to explain portfolio weights. After controlling for firm fixed effects, the results indicate that fund managers allocate significantly more capital to securities of both *FullyVisible* and *FullyHidden* friends, with the latter effect being 2.5 times larger. These findings are economically important. In conclusion, the different specifications indicate a consistent pattern: Fund managers invest more heavily in their friends' securities, and these allocations appear to be highly dependent on the visibility of their friendships.[13]

2.2. Performance

Our findings thus far indicate that fund managers allocate significantly more capital to firms managed by hiddenly-connected firm officer Facebook friends. We now explore whether this trend is associated with superior performance; that is, if the funds' holdings in the fund managers' hiddenly-connected stocks outperform their holdings in visibly connected and non-connected stocks. If fund managers benefit from managing the visibility of their friendships, we would expect to observe positive returns in stocks associated with such connections. In contrast, if fund managers allocate money to these stocks for other reasons, such as familiarity, we would anticipate nonpositive results. To investigate this question, we use a conventional calendar-time portfolio approach similar to Coval and Moskowitz's (2001) to create replicating portfolios based on the funds' holdings. In this process, we sort the funds' connected stocks into portfolios depending on their *FriendshipVisibility* levels. For the sake of brevity in the following descriptions, we subsume the various expressions of this variable under the term "connected". For each fund-month observation, we assign the stocks in a fund's portfolio into subportfolios based on whether any of the fund's managers maintain a Facebook friendship with any of the firm's same-month officers.[14] We calculate monthly portfolio returns for each fund under the assumption that the funds did not change their holdings between two reporting dates as

$$R^{N}_{i,t} = \sum_{k\epsilon\mathcal{N}} \left(\frac{w_{i,k,t}}{\sum_{k\epsilon\mathcal{N}} w_{i,k,t}} \right) r_{k,t+1} \tag{2}$$

and

$$R^{O}_{i,t} = \sum_{k\epsilon\mathcal{O}} \left(\frac{w_{i,k,t}}{\sum_{k\epsilon\mathcal{O}} w_{i,k,t}} \right) r_{k,t+1} \tag{3}$$

where $\mathcal{N}$ is the set of stocks of a firm with an officer connected to at least one of fund i's fund managers, and $\mathcal{O}$ is the set of non-connected stocks in fund i's portfolio. Following stock assignments into "Connected" and "Non-Connected" subportfolios, we keep stocks in the subportfolios until the next reporting date, when the portfolios are rebalanced to reflect changes in holdings. We weight stocks by their dollar values in the fund's respective subportfolio. We then compute value-weighted averages of the returns in Eqs. (2) and (3) across funds at time t, weighting each fund portfolio return by the fund's total net asset value in the subportfolio. This approach assigns greater weight to funds that have larger dollar values of investments, and effectively corresponds to a simple investment strategy of investing in the entirety of connected and non-connected stocks in proportion to the amounts held by our sample of funds.[15] The sample averages 169 Facebook-connected funds per month, with each fund each month on average holding two connected stocks and 146 non-connected stocks. To assess the performance of the calendar-time portfolios, we employ three distinct measures: simple raw returns, risk-adjusted returns, and characteristics-adjusted returns. Risk-adjusted returns are computed based on the four-factor model of Carhart (1997), using the intercept from a regression of monthly excess returns on explanatory variables that include the monthly returns from the three (Fama and French, 1993) factor-mimicking portfolios and Carhart's (1997) momentum factor; the data are obtained from Ken French's data library.[16] To address potential biases highlighted in prior research (e.g., Cremers et al. (2013)), we also calculate characteristics-adjusted returns following the methodology of Daniel et al. (1997), referred to hereafter as DGTW-adjusted returns. Specifically, DGTW-adjusted returns are defined as a stock's raw return minus the return on a value-weighted benchmark portfolio of all CRSP firms in the same size, book-to-market, and one-year past return quintile.

Table 6 presents our primary results, emphasizing the role of *FriendshipVisibility* in determining the performance differences between connected and non-connected stock portfolios. Across all three performance metrics—raw returns, risk-adjusted returns, and DGTW-adjusted returns—the degree of *FriendshipVisibility* significantly amplifies the outperformance of connected stocks over their non-connected counterparts.

[12] The variation in this test arises from fund managers trading in and out of stocks with different *FriendshipVisibility* characteristics. Furthermore, turnover among fund managers or firm officers also contributes: when a fund manager leaves, or when a connected firm officer steps down and is replaced by someone not connected on Facebook, the connection typically dissolves and the stock is no longer considered connected.

[13] In specifications not reported, we control for industry fixed effects (based on the 48-industry classification used in Fama and French (1997)) and fund fixed effects (using the fund's Morningstar categories), both leading to more pronounced results than the specification in column 8.

[14] A given fund-stock-month observation may be assigned to multiple *FriendshipVisibility* portfolios when individual fund managers of a team-managed fund have connections to firm officers at the same firm that differ in visibility. The same applies when a single fund manager is connected to multiple firm officers at the same firm. In all cases, we evaluate each fund manager–firm officer pair separately, allowing fund-stock-month observations to appear in more than one *FriendshipVisibility* portfolio within the same month.

[15] When applying equal-weighting schemes instead of value-weighting schemes, the results in this section yield qualitatively similar patterns, with a general tendency toward more pronounced return patterns.

[16] Throughout the remainder of the paper, all references to "four-factor alpha" or "risk-adjusted return" use this same specification, estimated at a monthly or daily frequency depending on the underlying data.

Table 5
Portfolio weights in connected stocks by visibility.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	74.59***	74.59***	74.59***	74.59***	1.42	1.43	−24.80***	−24.83***
	[3.21]	[3.21]	[3.21]	[3.21]	[2.82]	[2.83]	[4.05]	[4.06]
FullyVisible	35.18***				1.05		8.68*	
	[7.06]				[1.71]		[5.10]	
FirmOfcrHides		63.88***						
		[8.67]						
FundMgrHides			95.80***					
			[13.22]					
FullyHidden				136.47***		50.94***		21.64***
				[15.34]		[7.57]		[5.46]
Controls	No	No	No	No	Yes	Yes	Yes	Yes
Fixed Effects	Period	Period	Period	Period	Period Fund	Period Fund	Period Firm	Period Firm
Adj. R Squared	0.01	0.01	0.01	0.01	0.36	0.36	0.40	0.40
N ($\times 10^6$)	34.49	34.49	34.48	34.47	31.28	31.27	31.28	31.27

This table reports coefficient estimates and standard errors (in basis points) from Panel OLS regressions of funds' portfolio weights. The dependent variable w represents the fund's dollar investment in a stock as a percentage of the fund's total net assets. The independent variables measure the level of *FriendshipVisibility*, defined by a set of four dummy variables capturing the degree of visibility of a Facebook friendship between a fund manager and a firm officer depending on whether the friendship is publicly observable through both the fund manager's and the firm officer's friends lists (*FullyVisible*); observable through the fund manager's friends list but not through the firm officer's friends list (*FirmOfcrHides*); not observable through the fund manager's friends list but observable through the backlink from the firm officer's friends list (*FundMgrHides*); or neither observable through the fund manager's friends list nor through the firm officer's friends list (*FullyHidden*). Control variables, where included, are $Style$, defined as the percentage of the fund's total net assets allocated to the style of the stock under consideration (style is calculated as in Daniel et al. (1997)), and pME, pBM, and $R12$, which are percentiles of market value of equity, book to market, and past 12-month return, respectively. Each regression includes period fixed effects. Fund and firm fixed effects are included where indicated. The sample spans 1984–2020. Unit of observation is stock-fund-period. Standard errors are clustered at the fund and period levels and reported in brackets. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 6
Returns on connected stocks by visibility.

	Raw Return			Four-Factor Alpha			DGTW-Adjusted		
	Connected	Non-Conn.	LS	Connected	Non-Conn.	LS	Connected	Non-Conn.	LS
FullyVisible	1.05***	0.98***	0.07	0.11	−0.00	0.11	0.15	0.03	0.12
	(3.07)	(3.92)	(0.28)	(0.43)	(−0.14)	(0.44)	(0.70)	(0.63)	(0.57)
FirmOfcrHides	1.18***	0.98***	0.20	0.30	−0.01	0.32	0.34	0.02	0.32
	(3.42)	(3.72)	(0.74)	(1.12)	(−0.43)	(1.16)	(1.62)	(0.41)	(1.53)
FundMgrHides	1.53***	0.95***	0.57*	0.56*	−0.00	0.56*	0.71***	0.03	0.69***
	(3.73)	(3.99)	(1.87)	(1.81)	(−0.02)	(1.83)	(2.80)	(0.67)	(2.74)
FullyHidden	2.40***	0.93***	1.48***	1.35***	−0.00	1.36***	1.39***	0.03	1.37***
	(5.02)	(3.62)	(3.88)	(3.54)	(−0.16)	(3.57)	(3.84)	(0.57)	(3.80)

This table reports monthly calendar-time portfolio returns sorted by levels of *FriendshipVisibility*, which is a vector of four dummy variables capturing the degree of visibility of a Facebook friendship observed between a fund manager and a firm officer depending on whether the friendship is publicly observable through both the fund manager's and the firm officer's friends lists (*FullyVisible*); observable through the fund manager's friends list but not through the firm officer's friends list (*FirmOfcrHides*); not observable through the fund manager's friends list but observable through the backlink from the firm officer's friends list (*FundMgrHides*); or neither observable through the fund manager's friends list nor through the firm officer's friends list (*FullyHidden*). For each fund-month observation, we sort the stocks in each fund portfolio into "Connected" and "Non-Connected" portfolios. "Connected" stocks are defined as firms headed by one of the fund manager's then-active firm officer friends. Assuming that funds do not modify their holdings between two reporting dates, we construct monthly portfolios by retaining the stocks in the portfolios until the next reporting date, when they are rebalanced to reflect changes in holdings. Within a given portfolio, stock returns are weighted by the fund's dollar holdings. We then compute value-weighted returns by aggregating across funds, weighting each fund portfolio return by the fund's total net asset value in the subportfolio. Long-Short (LS) is the monthly return of a zero-cost strategy that buys the Connected portfolio and shorts the Non-Connected portfolio. We report raw returns, four-factor alphas, and DGTW-adjusted returns in the period 1984–2020. t-statistics appear in brackets beneath each estimate. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Notably, stocks classified in the *FullyHidden* category demonstrate the most substantial abnormal returns, achieving a monthly four-factor alpha of 135 basis points (t-stat = 3.54). Furthermore, a long-short strategy that involves purchasing *FullyHidden* stocks and selling short the fund managers' non-connected stocks generates an average monthly four-factor alpha of 136 basis points that is highly statistically significant (t-stat = 3.57), underscoring the considerable performance advantage associated with hidden connections. Stocks in the *FundMgrHides* category also exhibit positive abnormal returns, though to a lesser extent, with a monthly four-factor alpha of 56 basis points (t-stat = 1.81). Similarly, a long-short strategy involving *FundMgrHides* stocks yields an alpha of comparable magnitude. In contrast, trading *FirmOfcrHides* and *FullyVisible* stocks results in positive but statistically insignificant four-factor alphas of 30 and eleven basis points, respectively. The findings are consistent across different performance metrics, including DGTW-adjusted returns. In summary, our results indicate that fund managers' investment ideas in connected stocks experience very different fates: those associated with *FullyHidden* connections generate consistent, large, and significant risk-adjusted returns that outperform trades in visibly connected and non-connected stocks, underscoring the critical role of hidden connections in driving superior investment performance.

In addition to assessing the performance of hiddenly-connected stocks based on the funds' holdings, we also examine the fund managers' informational advantage based on their trading activity, offering further insight into both buy-side and sell-side behavior. To infer trades from holdings, we follow a method similar to Pool et al. (2015), which zeroes out periods of non-trading in a stock from the return series.

Table 7
Trade-based returns on connected stocks.

	Connected			Non-Connected			
	(1) Buy	(2) Sell	(3) LS	(4) Buy	(5) Sell	(6) LS	(3)–(6) Spread
FullyVisible							
All Trades	0.32	−0.18	0.46	0.05	0.16	−0.11	0.57
	(0.54)	(−0.31)	(0.55)	(0.37)	(1.26)	(−0.77)	(0.67)
Extensive Trades	0.30	−0.27	0.60	0.15	0.17	−0.04	0.64
	(0.35)	(−0.30)	(0.57)	(0.97)	(1.00)	(−0.20)	(0.60)
Intensive Trades	0.35	−0.11	0.46	0.08	0.16	−0.09	0.55
	(0.61)	(−0.25)	(0.66)	(0.52)	(1.18)	(−0.56)	(0.75)
FirmOfcrHides							
All Trades	0.48	−0.23	0.67	0.02	0.11	−0.11	0.78
	(0.94)	(−0.45)	(1.02)	(0.12)	(0.92)	(−0.75)	(1.17)
Extensive Trades	0.40	−0.56	0.80	0.18	0.25	0.05	0.75
	(0.51)	(−0.70)	(0.81)	(0.96)	(1.62)	(0.23)	(0.79)
Intensive Trades	0.51	−0.19	0.69	0.02	0.00	−0.02	0.71
	(0.96)	(−0.39)	(1.10)	(0.12)	(0.03)	(−0.12)	(1.07)
FundMgrHides							
All Trades	0.77**	−0.61*	1.36***	−0.09	−0.10	−0.01	1.36***
	(2.14)	(−1.94)	(3.15)	(−1.09)	(−0.91)	(−0.04)	(3.01)
Extensive Trades	1.45**	−1.04*	2.29***	−0.06	−0.05	0.12	2.17***
	(2.00)	(−1.85)	(3.06)	(−0.43)	(−0.38)	(0.66)	(2.86)
Intensive Trades	0.69*	−0.28	0.95**	−0.05	−0.09	0.03	0.92**
	(1.81)	(−0.89)	(2.10)	(−0.55)	(−0.98)	(0.30)	(2.02)
FullyHidden							
All Trades	1.27***	−0.68	1.89***	−0.04	0.01	−0.04	1.93***
	(2.75)	(−1.34)	(2.83)	(−0.34)	(0.06)	(−0.31)	(2.89)
Extensive Trades	2.25**	−1.90*	2.76***	−0.11	−0.21	0.02	2.74***
	(2.21)	(−1.84)	(2.79)	(−0.71)	(−1.27)	(0.14)	(2.78)
Intensive Trades	1.12**	−0.53	1.56**	0.02	0.15	−0.13	1.68***
	(2.42)	(−1.20)	(2.58)	(0.18)	(1.51)	(−0.96)	(2.75)

This table reports monthly trade-based portfolio returns sorted by levels of *FriendshipVisibility*, which include the expressions *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, and *FullyHidden*. To construct the portfolios, we assign connected and non-connected stocks into "Buy" and "Sell" portfolios depending on whether a stock was bought or sold compared to the previous fund-month observation. Within a given subportfolio, stocks are assigned the new dollars allocated relative to the previous month, and stock returns are weighted using the fund's total new dollar holdings in the subportfolio. We then compute value-weighted returns by aggregating across funds, using their total new subportfolio dollars as weights. Additionally, we differentiate Buy and Sell transactions into extensive (new or fully divested positions) and intensive (increased or decreased holdings) margin trades. Long-Short (LS) is the monthly return of a zero-cost strategy that buys the Buy portfolio and shorts the Sell portfolio. We report monthly four-factor alphas in the period 1984–2020. t-statistics appear in brackets beneath each estimate. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Specifically, for each fund-month observation, we assign connected and non-connected stocks into "Buy" and "Sell" portfolios depending on whether a stock was bought or sold compared to the previous fund-month observation. The funds' initial stock transactions are excluded from this process. Within a given subportfolio, stocks are assigned the new dollars allocated relative to the previous month, and stock returns are weighted using the fund's total new dollar holdings in the subportfolio. We then compute value-weighted returns by aggregating across funds, using their total new subportfolio dollars as weights. We further expand our analysis by decomposing "Buy" and "Sell" transactions into extensive and intensive margin trades, depending on whether—relative to the previous month—a fund's position in a stock is newly established ("extensive margin buy"), entirely sold off ("extensive margin sell"), expanded ("intensive margin buy"), or reduced ("intensive margin sell").

We present the results of our trade-based evaluation of portfolio returns in Table 7. For brevity, we focus on the four-factor alphas, noting that the DGTW-adjusted returns look very similar. Aligning with our earlier findings, the *FullyHidden* portfolio remains the most notable, yielding a monthly four-factor alpha of 127 basis points (t-stat = 2.75) on the buy side (column 1) and −68 basis points (t-stat = −1.34) on the sell side (column 2). Implementing a long-short strategy that goes long the Buy portfolio and short the Sell portfolio delivers a monthly four-factor alpha of 189 basis points (t-stat = 2.83), as reported in column 3. The observed spread between the long and short legs is plausibly attributable to mutual funds' short-selling constraints. Further, segmenting the trades into intensive and extensive categories reveals that intensive buys yield 112 basis points, and extensive buys 225 basis points per month. By contrast, intensive sells result in −53 basis points, and extensive sells −190 basis points per month. Notably, the performance ratio of extensive to intensive trades on the sell side is much higher (approximately 3.5 to one) compared to the buy side (roughly two to one). The stark variation in these ratios may signal underlying differences in how fund managers can respond to buy and sell ideas. When considering a stock favorably, a fund manager may establish a new position if the stock is not yet held or increase the position's size if it is. Conversely, when viewing a stock unfavorably, closing a position may reflect a stronger conviction compared to merely reducing the size of the position, as the latter suggests a deliberate choice not to divest completely.

Next, we examine whether the managerial information that we appear to capture has implications for longer-term stock valuations. We do this by exploring the portfolios' connected legs using event-time returns. Specifically, we compute value-weighted cumulative abnormal returns for the first 18 months following a fund's purchase of a connected stock, sorting by *FriendshipVisibility*. Each event window begins when a fund first takes a position in a connected stock (and restarts if the position is later fully liquidated and re-initiated). Abnormal returns are defined as the fund's stock-level return in excess of the market return, and monthly abnormal returns are value-weighted by the inflation-adjusted dollar size of each position across all connected fund–stock pairs within a given *FriendshipVisibility* bucket. We report the cumulative sums of these value-weighted abnormal returns in Fig. 4.

Consistent with our main results in Table 6, we find abnormal performance to increase monotonically with the level of *FriendshipVisibility*. Over the course of 18 months, the portfolio of stocks formed

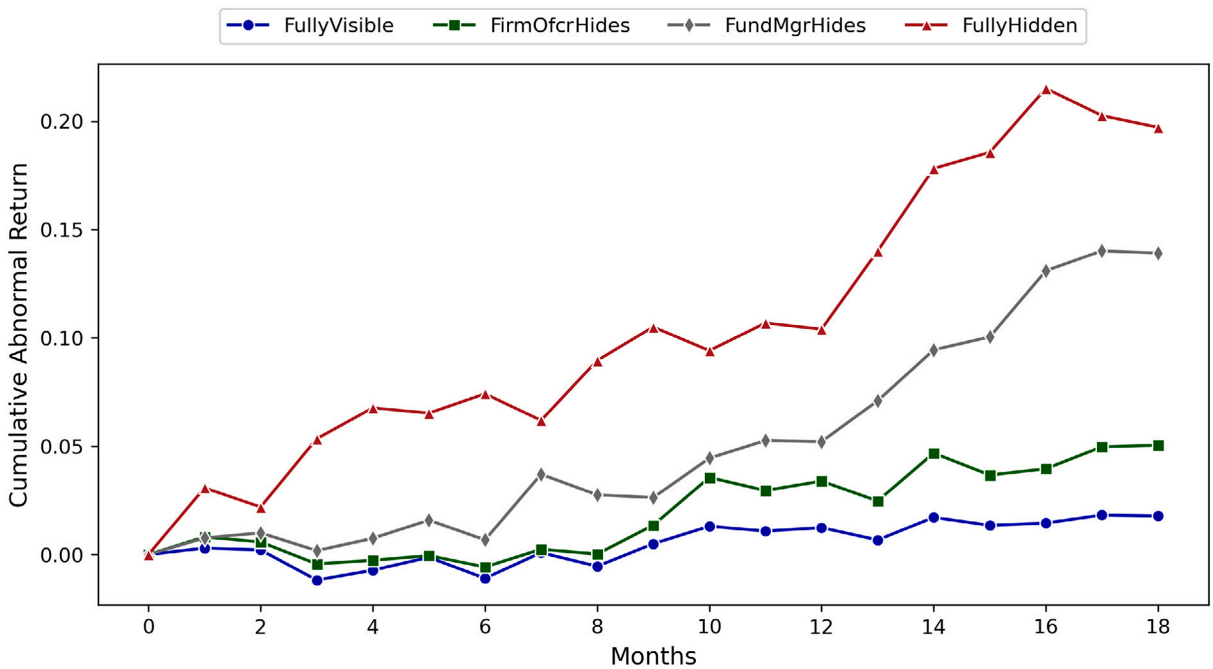

Fig. 4. CARs on connected stocks by friendship visibility.
This figure presents value-weighted cumulative abnormal returns (CARs) over the 18 months following a fund's purchase of a stock whose issuing firm includes a firm officer to which the fund manager maintains a Facebook-identified personal tie. The sample covers 1984–2020. Observations are grouped by levels of *FriendshipVisibility*, which include the expressions *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, and *FullyHidden*. Each event window starts with the initial purchase of a connected stock; if the position is fully liquidated and later repurchased, this repurchase initiates a new event window. Abnormal returns are calculated as fund-level stock returns net of market returns (based on the Fama–French market factor), and stock values are adjusted for inflation to keep the layered purchase events comparable in real terms. At each event month, monthly returns are value-weighted by the inflation-adjusted dollar size of each position across all connected fund–stock pairs within each visibility bucket. The plotted lines trace the cumulative sums of these value-weighted abnormal returns.

based on *FullyHidden* connections does not fall below the portfolios formed based on partially hidden connections—i.e., *FundMgrHides* or *FirmOfcrHides*—which in turn does not fall below the portfolio of stocks formed based on *FullyVisible* connections. The figure further indicates that the returns accrue gradually in the months after the initial purchase. In addition, it appears that the information we capture is eventually incorporated into stock prices and does not subsequently reverse.

2.3. Connected Not-Held portfolios

Motivated by the results presented in the previous section, we further analyze the relation between the fund managers' hiddenly-connected securities and performance. Given that mutual funds are limited in their ability to engage in short selling, the funds' active portfolio allocations may not accurately reflect the fund managers' full informational advantage. If positive information is reflected in the connected stocks held by the funds, we would expect negative information to be evident in the performance of the fund managers' connected stocks when they are not held by the funds. To gain further insights into this hypothesis, employing a similar portfolio construction approach to the one used in Section 2.2, we investigate returns on connected stocks that fund managers choose not to hold. In particular, for each fund-month observation, we sort the stocks in each fund portfolio into "Connected Held"[17] and "Connected Not-Held" portfolios. We place stocks in the Connected Not-Held portfolio if they are headed by a fund manager's then-active firm officer Facebook friend and are not held by the particular fund while being held by at least one other fund in the same Morningstar category in the given month. Assuming that funds do not modify their holdings between two reporting dates, we construct monthly portfolios by retaining the not-held stocks in the portfolio until the next reporting date, when the portfolios are rebalanced to reflect changes in holdings. Within a given portfolio, we weight a stock's returns by the stock's respective market capitalization. We then compute value-weighted returns by aggregating across funds, weighting each fund portfolio return by the fund's total net asset value in the subportfolio. The resulting sample comprises 2,613 distinct funds and 177,156 fund-month observations covering the period from January 1984 through December 2020. On average, a fund's Connected Not-Held portfolio per month includes 2.96 stocks managed by an average of 2.25 of the fund managers' then-active firm officer friends.

Table 8 presents the results of this analysis, indicating that the fund managers' hiddenly-connected stocks do not exhibit significant outperformance during the periods when they are not held by the funds. This observations holds true whether the assessment is based on four-factor alpha, shown in column 5, or DGTW-adjusted returns, shown in column 6. Additionally, we report the returns from a long-short strategy that involves selling short the Connected Not-Held portfolio while buying the Connected Held portfolio. For the *FullyHidden* portfolio, the set of connected stocks held by the funds outperforms the set of connected stocks not held by 119 basis points per month (t-stat = 2.59) in four-factor alpha, as shown in column 8, with DGTW-adjusted returns of a similar magnitude, as shown in column 9.

Stepping back, the Connected Not-Held results offer a more granular exploration of the same fund manager's decision to hold versus not hold hiddenly-connected stocks. This analysis provides evidence against a stock-specific selection narrative, as it involves a long-short strategy on the same set of hiddenly-connected stocks—focusing solely on the fund manager's active decision to hold or forgo these stocks.

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[17] In Section 2.3 we refer to the portfolio previously labeled "Connected" as "Connected Held" to distinguish it terminologically from the newly introduced "Connected Not-Held" portfolio, which contains stocks headed by the fund managers' firm officer friends that are currently not held by the fund.

Table 8
Returns on connected not-held stocks.

	Connected Held (CH)			Connected Not-Held (CNH)			Long CH/Short CNH		
	Raw	Alpha	DGTW	Raw	Alpha	DGTW	Raw	Alpha	DGTW
FullyVisible	1.05***	0.11	0.15	1.12***	0.06	0.05	0.05	0.11	0.17
	(3.07)	(0.43)	(0.70)	(3.09)	(0.30)	(0.28)	(0.15)	(0.35)	(0.63)
FirmOfcrHides	1.18***	0.30	0.34	1.18***	0.09	0.16	−0.09	0.10	0.08
	(3.42)	(1.12)	(1.62)	(3.44)	(0.56)	(0.97)	(−0.26)	(0.31)	(0.30)
FundMgrHides	1.53***	0.56*	0.71***	0.98***	−0.02	0.01	0.54	0.57*	0.71**
	(3.73)	(1.81)	(2.80)	(2.93)	(−0.09)	(0.05)	(1.59)	(1.66)	(2.57)
FullyHidden	2.40***	1.35***	1.39***	1.10**	0.16	0.20	1.30***	1.19***	1.20***
	(5.02)	(3.54)	(3.84)	(2.41)	(0.50)	(0.67)	(2.75)	(2.59)	(2.73)

This table reports monthly calendar-time portfolio returns for the funds' Connected Held and Connected Not-Held portfolios. Portfolios are sorted by levels of *FriendshipVisibility*, which include the expressions *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, and *FullyHidden*. For each fund-month observation, we sort the stocks in each fund portfolio into Connected Held (CH) and Connected Not-Held (CNH) portfolios. We place stocks in the Connected Not-Held portfolio if they are headed by a fund manager's then-active firm officer Facebook friend and are not held by the particular fund while being held by at least one other fund in the same Morningstar category in the given month. Assuming that funds do not modify their holdings between two reporting dates, we construct monthly portfolios by retaining the not-held stocks in the portfolio until the next reporting date, when the portfolios are rebalanced to reflect changes in holdings. Within a given portfolio, we weight a stock's returns by the stock's respective market capitalization. We then compute value-weighted returns by aggregating across funds, weighting each fund portfolio return by the fund's total net asset value in the subportfolio. Long CH/Short CNH is the monthly return of a zero cost portfolio that buys the CH portfolio and sells short the CNH portfolio. We report raw returns (Raw), four-factor alphas (Alpha), and DGTW-adjusted returns (DGTW) in the period 1984–2020. *t*-statistics appear in brackets beneath each estimate. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

2.4. Friendship intensity

If the systematic outperformance patterns we document are related to the varying visibility of social ties that we observe and measure, we would anticipate that more intense interactions among financial agents lead to stronger results. To explore this, we leverage a unique aspect of Facebook data, which offers advantages over the traditional network membership measures: the ability to assess not only the existence of connections but also the interactions within the network. Specifically, we examine detailed interaction activity—likes, comments, and tags—associated with our sample of individuals on the Facebook platform.

Among the 57.4 million interactions observed on our sample individuals' Facebook profiles—including user interactions with ten million uploaded photos (refer to Table 2 for details on the item-level Facebook data observed in this study)—we identify 73 thousand interactions exchanged directly between fund managers and firm officers. Of these, eleven thousand interactions occurred between pairs that we classify as "traded". Leveraging these observations, we compute portfolio returns on funds' connected stocks, sorted by an interaction indicator that captures whether the fund manager and firm officer associated with a given connected stock have interacted with each other's Facebook content (via likes, comments, or tags) within the one-year period preceding the fund's investment in the stock, or whether no such interaction was recorded. This analysis spans the period from February 2005 to December 2020, following Facebook's launch in February 2004.

Table 9 presents our results from the above test, shown across the various levels of *FriendshipVisibility*. For *FullyVisible* friendships, we observe modest monthly four-factor alphas of 15 basis points (t-stat = 0.41) with interactions and 13 basis points (t-stat = 0.39) without interactions, both of which are statistically insignificant. For *FirmOfcrHides* friendships, the four-factor alphas increase to 38 basis points in point estimate (t-stat = 1.30) with interactions and 33 basis points (t-stat = 1.19) without interactions, but again, these results are statistically indistinguishable from zero. For *FundMgrHides* friendships, we see four-factor alphas of 136 basis points (t-stat = 3.08) with interactions compared to 56 basis points (t-stat = 1.70) without. However, for stocks in the *FullyHidden* category, we again observe the strongest and most telling results: four-factor alphas of 235 basis points (t-stat = 3.80) with interactions and 129 basis points (t-stat = 2.56) without interactions. These findings suggest that while the intensity of relationships (indicated by interactions) influences performance across all categories, stocks in the *FullyHidden* category drive the largest and statistically most significant outperformance.

3. Indicators of informed trading

3.1. Returns around corporate news

Building on our previous analyses of returns earned on stocks associated with varying levels of *FriendshipVisibility*, we now seek to better understand the mechanisms driving the superior performance of connected stocks in the *FullyHidden* category. If the fund managers that are trading these stocks are better informed, we would expect the returns on these stocks to be more concentrated around realizations of firm-specific information into capital markets—such as corporate news announcements. By contrast, we would not expect to observe similarly pronounced return patterns for either non-connected stocks or connected not-held stocks. In the former case, the fund manager is less likely to have privileged access to information. The latter refers to instances where the fund manager is connected to a firm officer but elects not to hold the associated stock during the relevant period.

To investigate this further, we extend the portfolio construction methodology introduced in Sections 2.2 and 2.3. Specifically, we construct the Connected Held/Non-Connected Held and Connected Not-Held portfolios by assigning to each stock in each fund portfolio its daily return earned in the following month. For each fund-day observation, we sort the stocks in each fund portfolio into "News" and "No News" portfolios based on whether the given stock was the subject of a news announcement on that particular day. We weight stocks in the Connected Held/Non-Connected Held portfolios by their dollar values and those in the Connected Not-Held portfolios by their respective market capitalizations. We then compute value-weighted returns by aggregating across funds, weighting each fund portfolio return by the fund's total net asset value in the subportfolio.

To compile information on firm-specific news events, we use data available via RavenPack. The service provides collection and analysis of entity-related news using natural language processing and machine learning techniques. The RavenPack data are extracted from Dow Jones Newswires, The Wall Street Journal, FactSet, and tens of thousands of other traditional and social media sources. To ensure that the news items we use for our test convey material information about the firms rather than information about market movements, we follow Weller (2018) in excluding news reports on trading or prices (technical analysis signals, stock price movements, order imbalance reports) and on investor relations themes (typically announcements of future information revelation dates). We further filter the data down to include only news in which RavenPack considers the related firm to be playing a

Table 9
Returns on connected stocks by interaction intensity.

	Raw Return			Four-Factor Alpha			DGTW-Adjusted		
	Interacted	Non-Inter.	LS	Interacted	Non-Inter.	LS	Interacted	Non-Inter.	LS
FullyVisible	1.12***	1.07***	0.05	0.15	0.13	0.02	0.13	0.18	−0.05
	(2.86)	(3.05)	(0.25)	(0.41)	(0.39)	(0.28)	(0.29)	(0.34)	(−0.19)
FirmOfcrHides	1.29***	1.18***	0.12	0.38	0.33	0.05	0.39	0.37	0.02
	(2.99)	(3.41)	(0.33)	(1.30)	(1.19)	(0.51)	(1.18)	(1.12)	(0.06)
FundMgrHides	2.36***	1.47***	0.89*	1.36***	0.56*	0.81*	1.21***	0.68*	0.53*
	(4.03)	(3.61)	(1.87)	(3.08)	(1.70)	(1.82)	(2.67)	(1.85)	(1.69)
FullyHidden	3.42***	2.32***	1.11**	2.35***	1.29**	1.06**	2.46***	1.38***	1.09***
	(4.41)	(4.89)	(2.16)	(3.80)	(2.56)	(2.43)	(3.94)	(2.68)	(2.59)

This table reports monthly calendar-time portfolio returns on funds' connected stocks, sorted by an interaction dummy indicating whether the fund manager and firm officer associated with each connected stock have interacted with each other's Facebook content (through likes, comments, or tags) within the one-year period preceding the fund's investment in the stock ("Interacted"), or whether no such interaction was recorded ("Non-Interacted"). Additionally, portfolios are sorted by levels of *FriendshipVisibility*, which include the expressions *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, and *FullyHidden*. To construct the connected portfolios for this analysis, we use the portfolio construction approach detailed in Table 6. Long-Short (LS) is the monthly return of a zero-cost strategy that buys the Interacted portfolio and shorts the Non-Interacted portfolio. We report raw returns, four-factor alphas, and DGTW-adjusted returns in the period 2005–2020. *t*-statistics appear in brackets beneath each estimate. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 10
Returns on connected stocks around news.

	Connected Held (CH)		Non-Connected Held (NCH)		Connected Not-Held (CNH)		Long CH/ Short NCH		Long CH/ Short CNH	
	News	No News	News	No News	News	No News	News	No News	News	No News
FullyVisible	0.031*	0.006	0.019***	0.004	0.019**	−0.006	0.012	0.002	0.012	0.011
	(1.68)	(0.76)	(3.94)	(1.08)	(2.21)	(−0.75)	(1.02)	(0.24)	(1.04)	(1.21)
FirmOfcrHides	0.035*	0.006	0.019***	0.004	0.019**	−0.005	0.016	0.002	0.016	0.011
	(1.80)	(0.77)	(3.96)	(1.08)	(2.20)	(−0.72)	(1.32)	(0.25)	(1.44)	(1.17)
FundMgrHides	0.043**	0.007	0.019***	0.004	0.020***	−0.009	0.024**	0.002	0.023*	0.017*
	(2.06)	(0.79)	(3.92)	(1.08)	(2.69)	(−1.20)	(2.04)	(0.29)	(1.94)	(1.70)
FullyHidden	0.060***	0.005	0.020***	0.004	0.023*	−0.005	0.040***	0.001	0.037**	0.011
	(2.87)	(0.48)	(4.27)	(1.08)	(1.67)	(−0.44)	(2.69)	(0.08)	(2.39)	(0.81)

This table reports daily calendar-time portfolio returns on corporate news for the funds' Connected Held (CH), Non-Connected Held (NCH), and Connected Not-Held (CNH) portfolios. Portfolios are sorted by levels of *FriendshipVisibility*, which include the expressions *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, and *FullyHidden*. To construct the CH, NCH, and CNH portfolios for this analysis, we modify the portfolio construction approaches used in Tables 6 and 8 by assigning to each stock in each fund portfolio its daily return earned in the following month. For each fund-day observation, we sort the stocks in each fund portfolio into "News" and "No News" portfolios based on whether the given stock was the subject of a news announcement on that particular day. We weight stocks in the Connected Held/Non-Connected Held portfolios by their dollar values and those in the Connected Not-Held portfolios by their respective market capitalizations. We then compute value-weighted returns by aggregating across funds, weighting each fund portfolio return by the fund's total net asset value in the subportfolio. Long/Short represents the daily return of a zero-cost portfolio that buys the Connected Held portfolio and either sells short the Non-Connected Held portfolio or sells short the Connected Not-Held portfolio. We report daily four-factor alphas in the period 2000–2020. *t*-statistics appear in brackets beneath each estimate. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

key role (i.e., news items with an "event relevance" score of 100). To remove duplicate news reports, we isolate the first news item in chains of items that relate to the same subject (using RavenPack's "event similarity days" analytic). Following these preliminary data cleaning steps, we use the CUSIP bridge provided by RavenPack's entity mapping file to merge RavenPack's firm identifier (rp entity id) with CRSP's firm identifier (permco) and map the firms' news items to their respective stock returns. We align news items and stock returns using the New York Stock Exchange trading calendar. In this procedure, we follow a close-to-close rationale in accordance with CRSP's return formula.[18] Because the RavenPack data begin in 2000, the following analysis runs from January 2000 to December 2020.

Table 10 compares the average daily performance of the Connected Held, Non-Connected Held, and Connected Not-Held portfolios across all levels of *FriendshipVisibility* on days with and without news announcements. Initially, we observe that the Connected Held portfolio earns significantly positive returns around news announcements for all levels of *FriendshipVisibility*, with the most pronounced results in the *FullyHidden* specification, showing a daily four-factor alpha of six basis points (significant at the 1% level). In contrast, returns for the Connected Held portfolio on non-news days are small and statistically indistinguishable from zero, suggesting that the observed return premium is primarily generated on days with news headlines.

This pattern is also evident for stocks in the Non-Connected Held portfolio. Notably, a long-short strategy that buys connected stocks in the *FullyVisible* specification on news days and sells short non-connected stocks yields an insignificant daily four-factor alpha of 1.2 basis points, which more than triples to four basis points in point estimate (significant at the 1% level) in the *FullyHidden* specification. A similar trend is observed in the Connected Not-Held portfolio, with positive returns concentrated around news announcements. However, a long-short portfolio strategy that buys the Connected Held portfolio in the *FullyHidden* specification and sells short Connected Not-Held portfolio reveals that the Connected Held portfolio experiences significantly greater news returns on average than the Connected Not-Held portfolio, corroborating the evidence presented in Section 2.3.

[18] For example, if a news item becomes public during Friday evening after-market hours, we map it to the stock's Monday return to take into account that CRSP-reported daily stock returns are calculated based on a stock's closing price on a given day relative to the most recent valid closing price prior to this day.

Table 11
Returns by news type, expertise, and analyst coverage.

		FullyHidden	Non-Connected	Long-Short
Panel A: News Event Type				
Analyst Ratings		0.046	−0.009	0.055
		(0.53)	(−0.13)	(0.56)
Earnings and Revenues		0.234**	0.030	0.203**
		(2.27)	(0.21)	(2.26)
Capital Structure Changes		0.022	0.029	−0.007
		(0.21)	(1.21)	(−0.08)
Marketing and Investor Relations		0.076	0.008	0.068
		(1.24)	(0.40)	(1.22)
Mergers & Acquisitions		0.574***	0.184***	0.391***
		(4.80)	(6.98)	(3.37)
Other News		0.030	0.000	0.030
		(0.85)	(0.07)	(0.87)
Panel B: Firm Officer Expertise				
CEO	Earnings and Revenues	0.371	0.015	0.356
		(1.63)	(0.13)	(1.62)
	Mergers & Acquisitions	0.422*	0.103	0.319*
		(1.94)	(1.16)	(1.74)
CFO	Earnings and Revenues	0.683**	0.007	0.676**
		(2.31)	(0.09)	(2.27)
	Mergers & Acquisitions	1.083***	0.208	0.875**
		(2.59)	(1.38)	(2.00)
Other C-suite	Earnings and Revenues	0.068	0.010	0.058
		(0.59)	(0.18)	(0.56)
	Mergers & Acquisitions	0.108*	0.099	0.008
		(1.74)	(1.53)	(0.20)
Panel C: Analyst Coverage				
High Coverage		0.883*	−0.037	0.921**
		(1.90)	(−0.45)	(1.98)
Low Coverage		1.410***	−0.023	1.433***
		(3.51)	(−0.43)	(3.57)

This table reports daily four-factor alphas for portfolios of stocks held by connected fund managers, where the underlying connection falls into the *FullyHidden* specification, non-connected stocks, and the corresponding long–short spread. Panels A and B build on the portfolio construction approach in Table 10 by splitting returns based on firm-level news events. Panel A reports alphas by news category: Analyst Ratings, Earnings and Revenues, Capital Structure Changes, Marketing and Investor Relations, Mergers & Acquisitions, and Other News. Panel B focuses solely on Earnings and Revenues and M&A events, further broken out by selected executive roles—CEO, CFO, Other C-Suite—of the firm officer to whom the fund manager is connected. Panel C replicates the test from Table 6, dividing the *FullyHidden* portfolio by high versus low analyst coverage. We report daily four-factor alphas in the period 2000–2020. t-statistics appear in brackets beneath each estimate. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

3.2. Finer exploration of firm events

To further probe the nature of the informational advantage conveyed by hidden connections, we now examine which types of corporate events primarily drive the premia observed in the *FullyHidden* portfolios. Building on the methodology from Section 3.1, we classify firm-specific news according to six major event categories: (i) Earnings and Revenues, (ii) Mergers and Acquisitions (M&A), (iii) Capital Structure Changes, (iv) Marketing and Investor Relations, (v) Analyst Ratings, and (vi) Other News. This classification follows Jeon et al. (2022), and relies on RavenPack's event tagging framework. For each category, we compute daily four-factor alphas for three portfolios: stocks held by fund managers connected through *FullyHidden* ties, non-connected stocks, and the resulting long–short spread.

As reported in Panel A of Table 11, the strongest effects arise around earnings and M&A events. On earnings days, the long–short portfolio earns an economically meaningful and statistically significant alpha of 20 basis points (t-stat = 2.26). On M&A days, the effect is even more pronounced, with a daily alpha of 39 basis points (t-stat = 3.37). These patterns suggest that *FullyHidden* connections are particularly valuable during major corporate events where informational asymmetries are likely to be most pronounced. By contrast, return differentials are small and statistically insignificant on analyst rating days, capital structure announcements, and other residual news types.

3.3. Linking specific firm-role connections to firm events

Firm-level connections of fund managers may vary in the extent to which they provide informational advantages. We continue our focus on *FullyHidden* connections to examine whether the value of a hidden tie depends on the specific role of the connected firm officer. Certain positions—such as the CEO or CFO—may provide closer proximity to strategically or financially relevant information than others. To investigate this, we identify the role of the connected firm officer using the data available from BoardEx and sort returns by firm officer role around the two types of corporate events that previously exhibited the strongest return effects: earnings announcements and M&A activity (see Section 3.2).

Panel B of Table 11 shows that ties to CEOs and CFOs are especially informative. *FullyHidden* connections to CFOs, in particular, yield the highest long-short alphas around both event types—67 basis points on earnings days (t-stat = 2.27) and 88 basis points on M&A days (t-stat = 2.00). This aligns with CFOs playing a central role in financial disclosures and corporate transactions. Ties to CEOs also yield economically meaningful effects, though somewhat smaller in magnitude. By contrast, connections to other C-suite officers translate into more modest or insignificant return differentials. These results suggest that the value of hidden connections is shaped not only by whether the tie is concealed but also by the organizational role of the connected individual within the firm.

Table 12
Cross-sectional Fama–Macbeth regressions.

	FullyVisible	FirmOfcrHides	FundMgrHides	FullyHidden
Constant	0.0104***	0.0108***	0.0124***	0.0111***
	[0.0038]	[0.0038]	[0.0038]	[0.0039]
β(DiffWeight)	0.0142	0.0152*	0.0171**	0.0228***
	[0.0088]	[0.0089]	[0.0085]	[0.0083]
ME	−0.0006**	−0.0006**	−0.0008**	−0.0007**
	[0.0003]	[0.0003]	[0.0003]	[0.0003]
BM	0.0005	0.0005	0.0005*	0.0004
	[0.0003]	[0.0003]	[0.0003]	[0.0003]
MOM	0.0002	0.0003	0.0002	0.0002
	[0.0023]	[0.0023]	[0.0023]	[0.0024]
STR	−0.0238***	−0.0240***	−0.0230***	−0.0238***
	[0.0057]	[0.0057]	[0.0057]	[0.0058]
IMOM	0.0846***	0.0830***	0.0826***	0.0839***
	[0.0187]	[0.0186]	[0.0190]	[0.0190]
SUE	−0.0009	−0.0010	−0.0006	−0.0011
	[0.0012]	[0.0013]	[0.0012]	[0.0013]
Adj. R Squared	0.0680	0.0702	0.0783	0.0814
N	1,234,981	1,234,612	1,235,862	1,233,925
N Months	290	290	289	288

This table reports risk premium estimates from monthly cross-sectional (Fama and MacBeth, 1973) regressions for the period 1984–2020. The main independent variable of interest is *β(DiffWeight)*, estimated from a rolling regression of $r_{k,t}$ on $DiffWeight_{k,t-1}$. $DiffWeight_{k,t}$ is the difference between the average weight that Facebook-connected funds allocate to a given stock and the average weight that all other funds allocate to the stock. Other independent variables include firm size *(ME)*, book-to-market ratio *(BM)*, momentum *(MOM)*, short-term reversal *(STR)*, industry momentum *(IMOM)*, and standardized unexpected earnings *(SUE)*. The dependent variable in the Fama-MacBeth regressions is next month's stock excess returns *(ExcessRet)*, calculated as raw return minus the risk-free rate. All dependent and independent variables are winsorized at the 1st and 99th percentile each month. Regressions are run separately for levels of *FriendshipVisibility*, which include the categories *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, and *FullyHidden*. Standard errors are adjusted for clustering at the period level and are reported in brackets. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

3.4. Analyst coverage

To examine whether the effect of hidden connections depends on the availability of external information, we examine variation by analyst coverage. The underlying logic is that if fully hidden ties convey incremental information, their return effects should be most pronounced when alternative sources are scarce. We obtain quarterly analyst coverage from IBES and compute the median number of analysts covering each stock in each month. Using the global median across all firm-months as a cutoff, we split the *FullyHidden* portfolio into high- and low-coverage subgroups. Panel C of Table 11 shows that the effect is indeed stronger in low-coverage environments: the long-short return spread is 143 bps per month (t-stat = 3.57) in the low-coverage group compared with 92 bps (t-stat = 1.98) in the high-coverage group. While the magnitude remains economically meaningful even under greater analyst scrutiny, the difference is consistent with hidden ties being most valuable when traditional information channels are weaker.

3.5. SEC insider trading actions

Given the economic significance of hidden connections—particularly those classified as *FullyHidden*—and the abnormal returns associated with them, we examine whether such connections are also linked to regulatory enforcement. To this end, we compile the full universe of administrative proceedings and litigation releases published by the U.S. Securities and Exchange Commission (SEC) between 1996 and 2024, encompassing over 29,800 individual cases.[19] We match this enforcement dataset to our sample of fund managers using names and additional identifying details to detect any individuals named in insider trading investigations or legal actions. Using this matched sample, we find that fund managers subject to SEC enforcement actions are significantly more likely to conceal their Facebook friendships. Specifically, 71% of managers facing SEC charges hide their Facebook friends lists, compared to 44% among the broader manager population—an increase of over 60% that is statistically significant (t-stat = 3.14). These findings further support our interpretation that Facebook connection concealment is systematically associated with elevated insider trading risk.

To further contextualize our findings, we build on Ahern (2017), who documents that insider information frequently travels through strong social ties, and that such ties are associated with sizable abnormal trading gains. The underlying dataset, constructed from enforcement filings involving individuals prosecuted by the SEC or the Department of Justice (DOJ), records 1,139 insider tips exchanged among 622 individuals during the period 1996–2013. These records were augmented by the author to include, where available, detailed biographical information such as full names, birth years, educational histories, graduation dates, occupational backgrounds, and places of residence. Drawing on this data, we first exclude observations in which the identities of either the tipper or the tippee in a co-conspirator pair are unknown. We then follow the procedure outlined in Section 1.2 to identify these individuals' Facebook profiles. After these data cleaning steps, 399 individuals remain, of whom we are able to identify 157 on Facebook.[20] We find that 64% of the pairs in which both tipper and tippee are Facebook-identified in Ahern (2017)'s data appear connected on the platform. More broadly, 72% of those identified on Facebook hide their friends lists, exhibiting concealment behavior strikingly similar to that observed in our main sample. This exercise provides valuable external validation from an enforcement setting, reinforcing our interpretation of *FriendshipVisibility* as a meaningful marker for insider trading risk—and more generally, as a proxy for information-sensitive relationship management in financially sensitive contexts.

4. Robustness

4.1. Alternative stock-level performance test

To assess the robustness of our main findings in Section 2, we complement the portfolio analysis with an alternative stock-level performance test. Specifically, we implement multivariate cross-sectional Fama-MacBeth regressions (Fama and MacBeth, 1973) to evaluate the persistence of abnormal returns associated with fund managers' hiddenly-connected stocks. This approach enables us to control for a range of firm- and stock-level characteristics that are known to influence returns and are widely used in the asset pricing literature. These control variables include firm size *(ME)*, book-to-market ratio *(BM)*, momentum *(MOM)*, short-term reversal *(STR)*, industry momentum *(IMOM)*, and standardized unexpected earnings *(SUE)*. The dependent variable in the Fama-MacBeth regressions is the stock's excess return in the subsequent month. We calculate our primary regressor, $DiffWeight_{k,t}$, for each month t and stock k as the difference between the average weight that Facebook-connected funds allocate to the stock and the average weight allocated by all other funds. To ensure comparability across models, we standardize $DiffWeight_{k,t}$ by dividing it by its cross-sectional standard deviation each month. We lag *DiffWeight* by one month when forming the rolling betas; thus β(*DiffWeight*) is estimated using information through t-1 and is priced against $r_{k,t}$.

Table 12 presents the regression coefficient estimates for the average risk premia. The sample covers 1.24 million stock-month observations across 290 months (1984–2020). Control variables load

[19] Data obtained from SEC Administrative Proceedings and SEC Litigation Releases.

[20] The lower identification rate relative to our main sample likely reflects the fact that individuals in Ahern (2017)'s sample are, on average, nine years older.

Table 13
Characteristics by Facebook identification and friends list visibility.

		Panel A					Panel B				
		FB-identified		Non-identified		Diff.	Hidden Friends		Public Friends		Diff.
		Mean	SD	Mean	SD		Mean	SD	Mean	SD	
Fund Managers	Age	44.43	3.53	47.67	2.29	−3.25***	43.52	3.91	44.98	3.41	−1.45*
	Tenure	4.59	1.24	5.37	1.11	−0.78***	4.32	1.27	4.77	1.25	−0.46
	Male	0.86	0.02	0.93	0.02	−0.06***	0.86	0.05	0.87	0.02	−0.01
	CFA	0.56	0.06	0.54	0.07	0.02	0.58	0.08	0.56	0.07	0.02
	MBA	0.61	0.05	0.59	0.04	0.02*	0.60	0.06	0.61	0.04	−0.01
	Ivy League	0.33	0.09	0.30	0.09	0.02	0.33	0.11	0.32	0.09	0.01
	Growth Fund	0.41	0.07	0.38	0.08	0.03	0.39	0.06	0.41	0.08	−0.02
	Value Fund	0.20	0.03	0.22	0.07	−0.02	0.21	0.05	0.20	0.05	0.01
	Small-Cap Fund	0.21	0.04	0.19	0.07	0.02	0.22	0.14	0.20	0.04	0.02
	Large-Cap Fund	0.50	0.03	0.52	0.08	−0.02	0.51	0.08	0.49	0.04	0.01
	Fund TNA (bn)	1.29	1.01	0.98	0.60	0.31	1.45	1.13	1.18	0.93	0.27
Firm Officers	Age	47.50	4.86	53.68	2.18	−6.18***	46.01	5.62	48.39	4.77	−2.38*
	Tenure	5.42	0.65	6.56	0.66	−1.14***	4.94	0.70	5.72	0.69	−0.78***
	Male	0.80	0.07	0.85	0.04	−0.05***	0.78	0.09	0.81	0.06	−0.04**
	MBA	0.28	0.04	0.27	0.01	0.01	0.26	0.04	0.28	0.05	−0.02
	Ivy League	0.23	0.07	0.21	0.06	0.02	0.23	0.10	0.23	0.07	0.00
	Firm Size (bn)	47.03	26.43	39.50	20.02	7.53	50.23	30.50	43.87	22.95	6.36

This table presents demographic and organizational characteristics of fund managers (upper panel) and firm officers (lower panel). Panel A compares individuals by Facebook identification status (FB-Identified vs. Non-Identified), and Panel B compares individuals with hidden versus public friends lists. Fund managers and firm officers are evaluated on shared characteristics including age, experience, tenure, gender, MBA, and Ivy League education. Fund manager characteristics additionally include CFA designation and total net assets (TNA) managed, while firm officer characteristics include firm size. Observations are weighted by fund-month for fund managers and by fund-month-stock for firm officers, covering data spanning from 1984 to 2020. The table reports means, standard deviations (SD), and group differences (Diff.), with significance assessed using t-tests. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

with the expected signs from the literature: smaller firms earn higher next-month returns, short-term reversals are negative, and industry momentum is strongly positive, with value and momentum loading with the correct sign, but insignificant during our sample. Corroborating our findings in Table 6, we observe that the outperformance on Facebook-connected stocks increases with the level of *FriendshipVisibility*. For connected stocks in the *FullyVisible* category, the coefficient estimate of β(*DiffWeight*) is insignificant. However, we find significant coefficients of 152 and 171 basis points for stocks in the *FundMgrHides* and *FirmOfcrHides* categories, respectively. For stocks in the *FullyHidden* category, the coefficient estimate indicates that a one-unit higher beta to *DiffWeight* predicts an increase in monthly excess returns of 228 basis points.

Overall, our multivariate Fama-MacBeth regression analyses provide corroborating evidence to the portfolio results. Namely, that now even controlling for known return determinants, hidden connections appear to result in significantly outperforming trades for fund managers. Moreover, these hidden connections afford a substantively larger return premium than publicly visible connections, underscoring the importance of concealed social ties in driving investment outcomes.

4.2. Comparison of fund manager and firm officer characteristics

Throughout the paper we examine the portfolio decisions of fund managers in light of their Facebook friendships with firm officers—paying particular attention to cases in which both individuals hide their friends list. Our analysis relies on the subset of individuals we can match on Facebook. Although we document robust patterns in trading behavior and information value, it remains possible that systematic differences exist between individuals identified on Facebook and those who remain unidentified. Across the full 1984–2020 sample we locate the Facebook profiles for 41.8% of the 10,036 distinct fund managers and for 36.1% of the 281,829 distinct firm officers in our benchmark universes (see Fig. 2). Table 13 uses these coverage groups in two ways: it first compares Facebook-identified and non-identified individuals, and—within the matched sample—contrasts those with hidden friends lists to those with public friends lists.

Panel A of Table 13 shows that Facebook-identified fund managers are largely similar to their non-identified peers across a range of demographic and portfolio characteristics. The only significant differences are that Facebook-identified fund managers are modestly younger on average (hence slightly shorter tenure) and include a higher share of females (roughly seven percentage points). Otherwise, the two groups are comparable in terms of portfolio sizes (TNA), prevalence of MBA, CFA, and Ivy League degrees, and the distribution of growth, value, small-cap, and large-cap funds. Fig. 5 corroborates this similarity: the value-weighted four-factor alphas of funds run by Facebook-identified and non-identified fund managers move almost one-for-one, with a performance correlation of 97%. However, a specific subset of stocks—those in the *FullyHidden* category—shows significant outperformance when held by Facebook-identified managers, highlighting an area of notable divergence. The same pattern of characteristics emerges for firm officers. Facebook-identified firm officers are slightly younger (with shorter tenure) and modestly less male than their non-identified counterparts, but they do not differ in MBA or Ivy-League incidence, and their firms are only marginally larger on average. All other measures between Facebook-identified and non-identified firm officers are comparable across the two groups.

Panel B of Table 13 turns to exploring the characteristics of fund managers and firm officers with hidden friends lists versus those with public friends lists. Fund managers with hidden friends lists are nearly identical to those with public friends across all demographic and portfolio characteristics. The lone marginal gap is in age—a difference of 1.45 years (t-stat = 1.71)—which does not translate into meaningful implications for tenure, experience, or investment mandate. In the remaining portfolio dimensions, the groups are statistically indistinguishable. Firm officers who hide their friends lists show the same small age and gender differences, but are otherwise equivalent to their public-list counterparts.[21]

Although fund managers and firm officers with hidden friends lists do not differ observably from their peers—and the funds' overall performance outside of connected stocks is not unusually strong—one might worry that friendships are formed or maintained more intensely after

[21] For those characteristics that are static across time (CFA, MBA, Gender, Ivy), we have also estimated this table at a purely cross-sectional level. The results are nearly identical; the only notable difference is that Facebook-identified fund managers and firms officers are slightly less likely to be male than those who are not Facebook-identified.

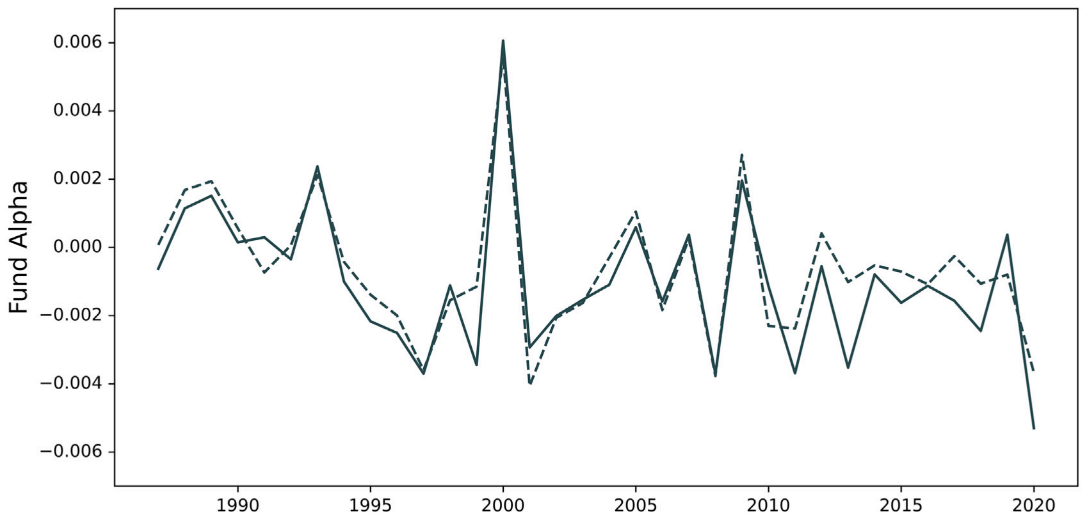

Fig. 5. Performance of Facebook-identified vs. non-identified funds.
This figure contrasts the annual value-weighted four-factor alpha of Facebook-identified funds (dashed line) versus all other, non-identified funds in the benchmark universe of mutual funds defined in Section 1.4 (solid line) over the 1984–2020 sample period. A fund is "Facebook-identified" if it is managed by a fund manager who was identified on Facebook. Each plotted point is obtained by first estimating, for each fund in each month, a four-factor alpha from a rolling 36-month regression (requiring at least 24 months of valid return observations) of the fund's excess return (net return minus the risk-free rate) on contemporaneous market excess return, SMB, HML, and momentum factors. These regressions use monthly net returns and total net assets (TNA) from MS Direct, together with factor returns from the Fama–French data library. The resulting monthly fund-level alpha is then weighted by the fund's one-month-lagged TNA to form two subportfolio aggregates (Facebook-identified vs. non-identified); these value-weighted alphas are summed to yield each subportfolio's monthly aggregate. Finally, for each calendar year, we take the simple average of its twelve monthly aggregate alphas to obtain that year's annual value-weighted alpha for each group. In the chart, the vertical axis measures annual alpha (in decimal form), and the horizontal axis measures calendar year; the dashed and solid lines trace the trajectories of Facebook-identified and non-identified funds, respectively.

good performance (reverse causality). Several pieces of evidence argue against this. First, Table 8 demonstrates that fund managers exhibit timing ability, suggesting that they know when to hold connected stocks poised to outperform and when to sell the others, pointing to an information-flow channel rather than an ex-post link. Second, Table 9 in Section 3.1 shows that returns are significantly larger following interactions with firm officers, indicating that hidden ties are being used contemporaneously rather than being ex-post friendship links. Together, these findings suggest that the abnormal returns associated with hidden Facebook friendships are not explained by observable demographics or simple reverse-causality stories.

4.3. Stronger results in post-inception period

Although Facebook was launched in 2004, the data that we observe extends back to 1984, enabling us to capture many long-standing relationships between fund managers and firm officers. For example, we locate a fund manager in a 1977 high-school graduation photograph alongside a friend who would later become the board chair of a firm whose stock she would eventually trade with notable success. In this sense, Facebook supplies a footprint that allows us to uncover and document these deeply rooted personal links. Nonetheless, a potential concern is that among the fund managers active in the earlier portion of our sample period (prior to Facebook's launch in 2004), we are only able to identify individuals who survived long enough and were sufficiently skilled to remain in the industry through 2010, 2015, and beyond—potentially introducing selection bias. However, Table 13 eases this worry, showing that Facebook-identified fund managers are indistinguishable from their peers on all observable characteristics. Moreover, throughout our analyses, we find no evidence that Facebook-identified fund managers outperform their peers on average. Rather, they perform curiously (and robustly) well on their hiddenly-connected stocks relative to their other holdings. In Table 14, we take this analysis one step further by conducting an additional robustness test that splits the sample into pre- and post-2004 sub-periods. This comparison is designed to assess whether look-ahead or survivorship effects might cause the earlier period to appear stronger. Instead, the opposite holds: the effects are more pronounced in the post-2004 sub-period, when our Facebook-identified coverage expands to include 40% of all fund managers. This finding is reassuring, as it empirically demonstrates that the strength of our results increases in tandem with the expansion of the sample's depth and breadth.

4.4. Replication with index funds

As a final robustness check, we conduct a falsification test using index funds. Naturally, our findings imply that the beneficial effects of hidden connections between fund managers and firm officers should be confined to actively managed funds, where managers exercise discretion in stock selection. In contrast, index funds—which follow passive investment strategies—should not exhibit performance variation related to *FriendshipVisibility*.

To test this implication, we replicate the core performance analysis from Table 6 using a sample of passively managed funds. We follow the same procedures described in Sections 1.2 through 1.6 to select U.S.-domiciled U.S.-equity index mutual funds available in MS Direct and to gather the data on fund managers, corresponding firm officers, and any friendship ties between them. We identify index funds by using the index fund flag in MS Direct and by scanning fund names for the terms "index" or "idx". This yields a sample of 434 index funds spanning the period 1984 and 2020, averaging 115 funds per calendar quarter and totaling 34,003 fund-month observations. These funds are managed by 677 distinct fund managers. Among these fund managers, we identify 284 individuals (41.9%) on Facebook, resulting in a final sample of 385 funds and 24,587 fund-month observations.

Using this dataset, we construct portfolios of connected and non-connected stocks, sorting these on levels of *FriendshipVisibility*. As shown in Table 15, we find no significant return differentials across those levels. Specifically, four-factor alphas are statistically insignificant for all expressions of this variable. The absence of performance

Table 14
Returns before and after Facebook's inception.

	Pre-Facebook four-factor alpha			Post-Facebook four-factor alpha		
	Connected	Non-Conn.	LS	Connected	Non-Conn.	LS
FullyVisible	0.13	0.04	0.08	0.09	−0.01	0.10
	(0.26)	(0.08)	(0.21)	(0.47)	(−0.18)	(0.49)
FirmOfcrHides	0.27	0.01	0.26	0.34	−0.01	0.36
	(0.68)	(0.35)	(0.67)	(0.82)	(−0.39)	(0.84)
FundMgrHides	0.49*	0.02	0.47*	0.79**	−0.01	0.81**
	(1.73)	(0.18)	(1.70)	(1.99)	(−0.20)	(2.03)
FullyHidden	1.21**	0.01	1.19**	1.57***	−0.01	1.58***
	(2.15)	(0.06)	(2.12)	(3.76)	(−0.12)	(3.79)

This table reports monthly calendar-time portfolio returns for connected and non-connected holdings of mutual funds, evaluated separately for the pre-Facebook period (1984–2003) and a post-Facebook period (2004–2020), split around Facebook's inception in February 2004. Portfolios are sorted by levels of *FriendshipVisibility*, which include the expressions *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, and *FullyHidden*. Portfolios are constructed according to the approach detailed in Table 6. Long-Short (LS) is the monthly return of a zero-cost strategy that buys the Connected portfolio and shorts the Non-Connected portfolio. We report four-factor alphas for each sub-sample. t-statistics appear in brackets beneath each estimate. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 15
Returns on connected stocks of index funds.

	Raw Return			Four-Factor Alpha		
	Connected	Non-Conn.	LS	Connected	Non-Conn.	LS
FullyVisible	0.87**	0.88***	−0.01	−0.03	0.03	−0.06
	(2.38)	(3.33)	(−0.09)	(−0.25)	(0.85)	(−0.12)
FirmOfcrHides	0.84**	0.82***	0.02	0.03	−0.00	0.03
	(2.34)	(2.97)	(0.12)	(0.31)	(−0.06)	(0.28)
FundMgrHides	0.84**	0.84***	0.00	0.01	0.02	−0.01
	(2.36)	(2.91)	(0.08)	(0.20)	(0.66)	(−0.09)
FullyHidden	0.88**	0.86***	0.02	0.02	0.02	0.00
	(2.41)	(2.79)	(0.13)	(0.19)	(0.52)	(0.11)

This table reports monthly calendar-time portfolio returns for connected and non-connected holdings of index funds. Portfolios are sorted by levels of *FriendshipVisibility*, which include the expressions *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, and *FullyHidden*. We construct the portfolios using the portfolio construction approach detailed in Table 6. Long-Short (LS) is the monthly return of a zero-cost strategy that buys the Connected portfolio and shorts the Non-Connected portfolio. We report raw returns and four-factor alphas in the period 1984–2020. t-statistics appear in brackets beneath each estimate. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

variation in index funds strengthens the interpretation that hidden connections confer informational advantages, but only when fund managers have the discretion to act on them.

5. Conclusion

This paper examines the trading performance associated with hidden connections between fund managers and firm officers. Such ties are suggestive of informed trading behavior and provide novel evidence consistent with insider trading. Our analysis draws on more than 100,000 manually identified Facebook profiles and their 35 million friendship ties. Leveraging unique features of the platform, we distinguish publicly observable friendships from those that are concealed. Trades linked to hidden connections earn the largest subsequent returns, whereas visible ties convey little predictive power. The endogenous behavioral choice to hide connections proves more informative than any previously documented, publicly observable network characteristic and emerges as the strongest single predictor of future stock returns.

Our results underscore that it is not merely the existence of social connections, but their deliberate obscurity, that shapes the investment decisions and outcomes we observe. The premium associated with hidden ties remains robust and significant to the present day and does not appear to reflect familiarity or selection biases. Instead, fund managers exhibit a nuanced awareness of when to engage with or avoid stocks linked to their concealed relationships. This effect persists across industries, firm types, time periods, and investment styles, underscoring its broad relevance.

By illuminating the hidden dimensions of financial networks, this paper advances understanding of how personal ties shape market behavior. It emphasizes the need to look beyond publicly observable links and to consider the concealment of relationships when assessing information flow, market efficiency, and compliance risk. Future research should probe the mechanisms through which hidden ties transmit information, the conditions that amplify their value, and the policy challenges they pose. More broadly, the findings highlight the interplay between strategic social behavior and market performance, offering insights for academics, policymakers, and investors alike.

CRediT authorship contribution statement

Manuel Ammann: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Alexander Cochardt:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lauren Cohen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Stephan Heller:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2025.104225.

Data availability

The mendeley data is available at https://data.mendeley.com/datasets/tjjf39mjng/2.

References

Ahern, Kenneth R., 2017. Information networks: Evidence from illegal insider trading tips. J. Finance Econ. 125 (1), 26–47.

Cai, Ye, Sevilir, Merih, 2012. Board connections and M&A transactions. J. Finance Econ. 103 (2), 327–349.

Carhart, Mark M., 1997. On persistence in mutual fund performance. J. Finance 52 (1), 57–82.

Chen, Huaizhi, Cohen, Lauren H., Gurun, Umit, Lou, Dong, Malloy, Christopher J., 2020. IQ from IP: Simplifying search in portfolio choice. J. Finance Econ. 138 (1), 118–137.

Cline, Brandon N., Posylnaya, Valeriya V., 2019. Illegal insider trading: Commission and SEC detection. J. Corp. Financ. 58, 247–269.

Cohen, Lauren H., Frazzini, Andrea, Malloy, Christopher J., 2008. The small world of investing: Board connections and mutual fund returns. J. Political Econ. 116 (5), 951–979.

Cohen, Lauren H., Frazzini, Andrea, Malloy, Christopher J., 2010. Sell-side school ties. J. Finance 65 (4), 1409–1437.

Coval, Joshua D., Moskowitz, Tobias J., 2001. The geography of investment: Informed trading and asset prices. J. Political Econ. 109 (4), 811–841.

Cremers, K. J. Martijn, Petajisto, Antti, Zitzewitz, Eric W., 2013. Should benchmark indices have alpha? Revisiting performance evaluation. Crit. Finance Rev. 2 (1), 1–98.

Daniel, Kent, Grinblatt, Mark, Titman, Sheridan, Wermers, Russ, 1997. Measuring mutual fund performance with characteristic-based benchmarks. J. Finance 52 (3), 1035–1058.

Elton, Edwin J., Gruber, Martin J., Blake, Christopher R., 2011. Holdings data, security returns, and the selection of superior mutual funds. J. Finance Quant. Anal. 46 (2), 341–367.

Engelberg, Joseph, Gao, Pengjie, Parsons, Christopher A., 2012. Friends with money. J. Finance Econ. 103 (1), 169–188.

Engelberg, Joseph, Gao, Pengjie, Parsons, Christopher A., 2013. The price of a CEO's rolodex. Rev. Finance Stud. 26 (1), 79–114.

Fama, Eugene F., French, Kenneth R., 1993. Common risk factors in the returns on stocks and bonds. J. Finance Econ. 33 (1), 3–56.

Fama, Eugene F., French, Kenneth R., 1997. Industry costs of equity. J. Finance Econ. 43 (2), 153–193.

Fama, Eugene F., MacBeth, James D., 1973. Risk, return, and equilibrium: Empirical tests. J. Political Econ. 81 (3), 607–636.

Jagolinzer, Alan D., Larcker, David F., Ormazabal, Gaizka, Taylor, Daniel J., 2020. Political connections and the informativeness of insider trades. J. Finance 75 (4), 1833–1876.

Jeon, Yoontae, McCurdy, Thomas H., Zhao, Xiaofei, 2022. News as sources of jumps in stock returns: Evidence from 21 million news articles for 9000 companies. J. Finance Econ. 145 (2), 1–17.

Massa, Massimo, Reuter, Jonathan, Zitzewitz, Eric W., 2010. When should firms share credit with employees? Evidence from anonymously managed mutual funds. J. Finance Econ. 95 (3), 400–424.

Patel, Saurin, Sarkissian, Sergei, 2017. To group or not to group? Evidence from mutual fund databases. J. Finance Quant. Anal. 52 (5), 1989–2021.

Pool, Veronika K., Stoffman, Noah, Yonker, Scott E., 2015. The people in your neighborhood: Social interactions and mutual fund portfolios. J. Finance 70 (6), 2679–2732.

Weller, Brian M., 2018. Does algorithmic trading reduce information acquisition? Rev. Finance Stud. 31 (6), 2184–2226.